CardioResp Monitor

User Guide for the web interface · Firmware 1.2 (final)

Heart-rate, breathing, chest-vibration and temperature monitor with a live dashboard on your phone, tablet or PC.

Covers: connecting, calibration, the five pages (Live, People, Mesh, Files, Settings), sensor placement and gravity, alarm sounds, PDF reports, colours and alerts, recording and downloads, licence, software update, mesh of several monitors, troubleshooting.

> **Not a medical device.** CardioResp is a research and wellness instrument. Its warnings are early hints that a person deserves attention; they are not diagnoses and never replace ECG, pulse oximetry, a thermometer or a clinician. In an emergency call your local emergency number first.

Screenshots in this guide were taken with demonstration data. The heart and breathing traces fill in on a live monitor.

Contents

1. What you need and how it works
2. Placing the sensor
3. Connecting to the dashboard
4. Start-up and calibration
5. Screen layout and navigation
6. Live page
7. Sensor placement and gravity
8. Detailed analysis and tools
9. Alarm sounds
10. Colours, severity and what to do
11. People page: patient and nurse
12. Mesh page (several patients)
13. Files page: recordings and PDF reports
14. Settings page (including About)
15. Licence and the basic (expired) mode
16. Heat, power modes and battery
17. Troubleshooting
18. Quick reference and glossary

1. What you need and how it works

In the box

- The **CardioResp monitor**: a small control unit with its chest sensor on a short cable.
- A USB power cable and adapter (or the built-in battery, if your monitor has one).
- A chest strap, medical tape or similar to hold the sensor firmly on the body.
- A phone, tablet or computer with a web browser (Chrome, Safari, Edge, Firefox).
- The card or label with the monitor's **Wi-Fi name and password**.
- If licensed: your **activation key** (a long text) from the supplier.

What the monitor measures

Quantity	How
Heart rate and rhythm	The sensor feels the heart's tiny mechanical vibration through the chest (seismocardiography, SCG) and its rotation (gyrocardiography, GCG).
Breathing rate and pattern	The chest tilts a little with every breath; the sensor measures that tilt.
Chest vibration	Strength of vibration (mg), rotation (°/s), share of high-frequency vibration (rises with coughing) and the number of cough-like bursts.
Temperature	The sensor's own temperature, shown in °C and °F, which becomes a body-temperature estimate after you calibrate it (chapter 14).
Alarms	A different sound for each kind of problem (heart, breathing pause, breathing, temperature, cough, sensor, connection), with an alarm bar on the screen (chapter 9).
Assessment	A rule-based severity (age-referenced limits, spectral and event rules) and a tiny machine-learning model run together and are written out as a short report.

The monitor creates its own Wi-Fi network (or joins your router) and shows the dashboard in your browser; no app is needed. Nothing is sent to the internet.

2. Placing the sensor

Placement matters more than anything else. A loosely held sensor is the most common reason for poor readings.

- **Best position:** the lower breastbone (sternum), on bare skin or a thin shirt, lying on the back or sitting.
- Also usable: the side of the ribcage or the back (the heart signal is weaker there).
- Hold the sensor **firmly** with a strap or tape. Strain-relieve the cable so it cannot tug the sensor.
- The patient should be **still and relaxed** during the first calibration and when you read values.

> **Check it on screen.** Orientation to use: the sensor's **Z axis out of the chest and Y axis towards the head**. The **Sensor placement and gravity** card on the Live page (chapter 7) shows where gravity points in the sensor's axes: lying flat on the back, the dot sits in the centre.

> **Tip.** If the body temperature reads poorly, place the sensor so that its flat face touches the skin and calibrate the temperature in the position you will use.

3. Connecting to the dashboard

The monitor can work in two ways and switches between them by itself.

	Home / clinic Wi-Fi	Direct connection
Phone joins	your normal Wi-Fi	the monitor's own network `CardioResp-XXXXXX`
Address	`http://<IP>` or `http://<name>.local`	`http://192.168.4.1`
Heat	cooler	warmer (the monitor runs its own network)

First connection (direct)

1. Switch the monitor on. Its Wi-Fi network is called `CardioResp-` followed by six characters (for example `CardioResp-531CD0`). The network name and password are on the label of the monitor or on the card supplied with it.
2. On your phone open Wi-Fi settings, join that network and enter the password.
3. The dashboard opens by itself (captive portal). If not, open `http://192.168.4.1` in the browser.

> **Note.** Your phone may say "no internet". Stay connected. Turn off mobile data if the phone keeps jumping to it.

Connecting the monitor to your home or clinic Wi-Fi

1. On first use the page shows the card **Connect the monitor to your Wi-Fi** at the top (it is also in Settings).
2. Press **Scan**, pick your network (2.4 GHz only), type the password. Optionally change the monitor's name (default `cardio-xxxxxx`, address `name.local`).
3. Press **Connect**. The phone may leave the monitor's network for a moment; rejoin it if asked.
4. On success the page shows the new address. Press **Keep this address fixed** so it never changes, and bookmark it.
5. The monitor's own network switches off after 2 minutes; join your normal Wi-Fi again.

If you do not want this, press **Not now, use the direct connection**. Every later start, the monitor tries the saved network for 15 seconds and falls back to its own network automatically.

Fixed IP address (optional)

Open **Fixed IP address (optional)** in the setup form, tick **Use a fixed address** and enter the IP address, gateway (router) and subnet mask. Choose an address outside the router's automatic range.

> **Security.** Anyone on the same network can open the dashboard. Use a trusted network. The monitor does not contact the internet.

4. Start-up and calibration

At every power-up the monitor calibrates for about 8 seconds. A banner at the top shows **Calibrating - keep still** with a progress bar. During this time:

- keep the sensor on the body and the patient **completely still**;
- the monitor stores the gravity direction and the gyroscope offsets and checks stillness;
- if there is movement it retries (up to 3 attempts).

"Gravity" here means the pull of the earth, always straight down. The accelerometer reports it as an **up direction in the sensor's own axes**. The direction measured during these 8 seconds is the **reference**: every later posture change and every breath is measured as a tilt away from it. The **Sensor placement and gravity** card on the Live page shows it (chapter 7).

Banner	Meaning	What to do
Calibrating - keep still	start-up calibration is running	stay still
Calibrated with a warning: the accelerometer reads 1.2 g	common with some sensors	nothing; the offset is corrected automatically. Optionally run the 6-position calibration (chapter 14)
Sensor not found	the sensor is not connected	check that the sensor cable is plugged in firmly; switch the monitor off and on; contact your supplier if it stays
Sensor connection lost	the sensor cable came loose	plug the cable in firmly; the monitor recovers by itself

After calibration the first heart rate appears after about 10 to 20 seconds. Heart-rate variability needs about 1 minute and the 5-minute breathing rate fills in over 5 minutes.

> **Tip.** After a large change of posture press **Settings → Calibration → Recalibrate sensor** and keep still for 8 seconds.

5. Screen layout and navigation

The top bar

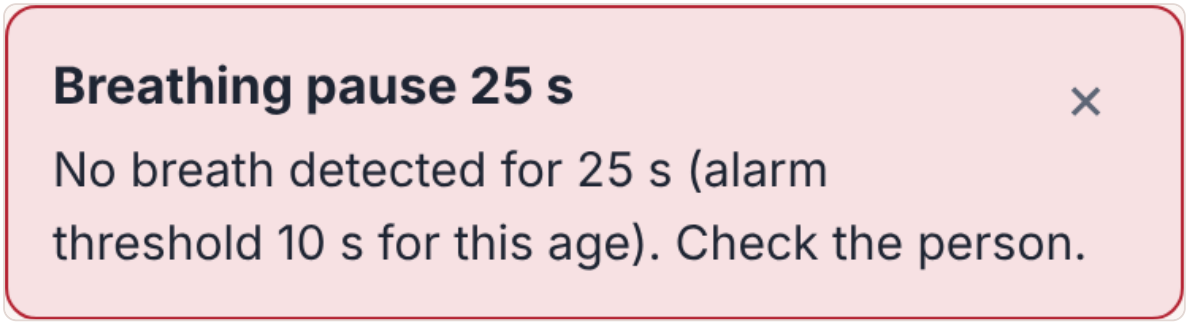


Top bar: monitor name, connection state, network mode, battery and temperature.

Item	Meaning
Pulse dot	flashes with each detected heart beat
CardioResp + name	this monitor's name (for example CardioA1B2)
Live / Reconnecting	green: live stream from the monitor. Red: connection lost. The page reconnects automatically and falls back to slower polling if needed.
Network chip	**Direct connection** or the home network name and signal
Battery	percentage and voltage, or "USB power"
Temperature	body/sensor temperature in °C and °F

Messages and how to close them

Messages appear under the top bar: calibration, alerts (for example "Breathing pause"), the licence notice and tips. Every message has a × in its top right corner. Tap it to close the message. It stays closed until its content changes: a **new** or **different** message appears again, so you never miss something new.

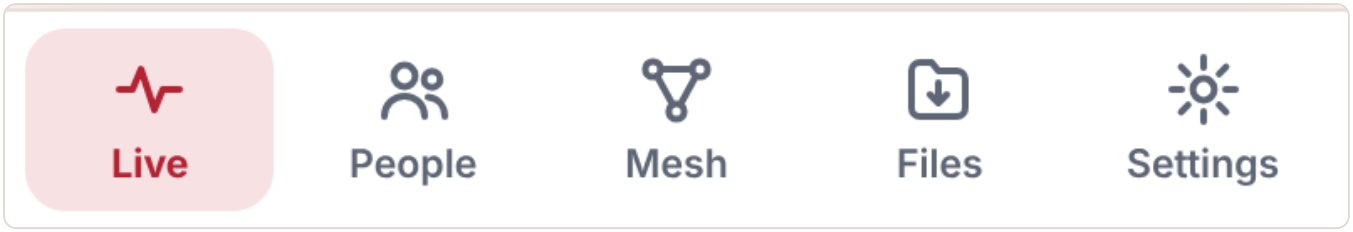


An alert message with its × close button.

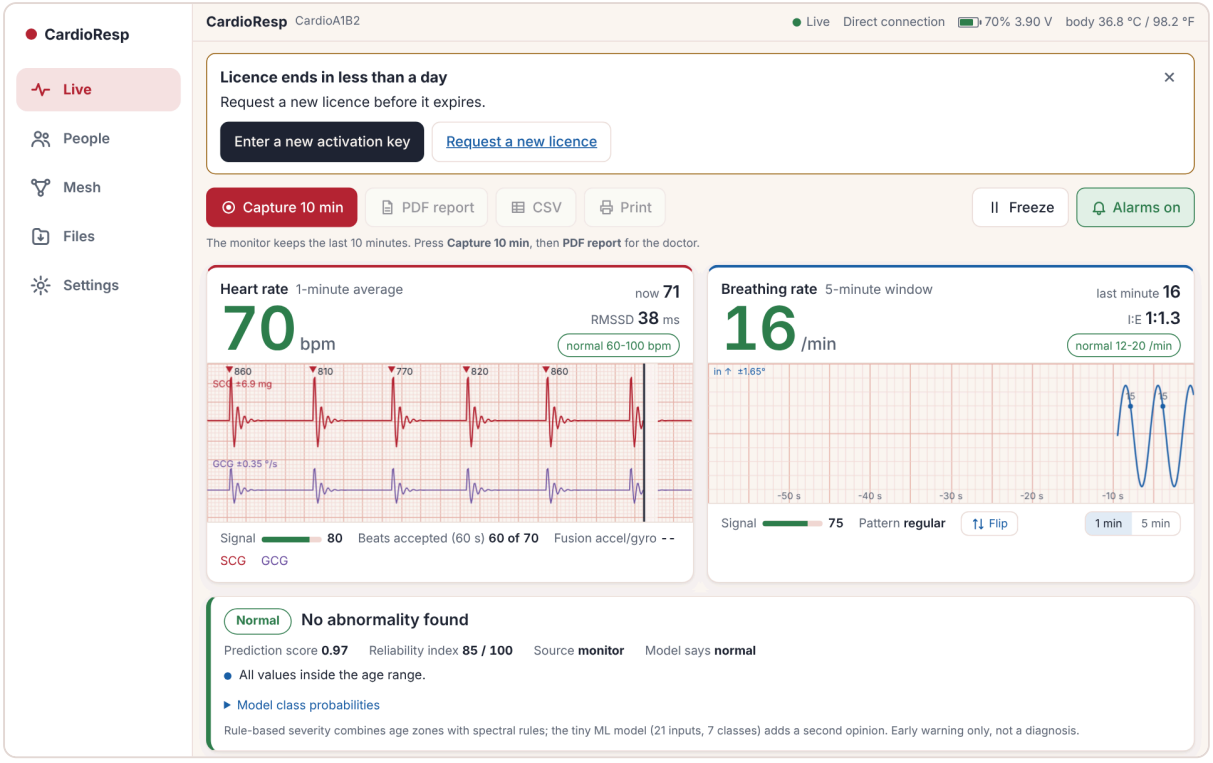
The alarm bar at the bottom also has a ×: it silences the current alarms for 2 minutes and hides the bar. A new alarm, or one that gets worse, brings the bar and the sound back (chapter 9).

The navigation bar

On a **phone** the five pages are always at the **bottom** of the screen, within reach of your thumb, so you never have to scroll back to the top to switch pages. On a **tablet or computer** they are in a bar on the **left side**. The current page is coloured (Live red, People blue, Mesh violet, Files green, Settings amber). A red dot on **Live** means an alarm is active.



Bottom navigation bar on a phone.



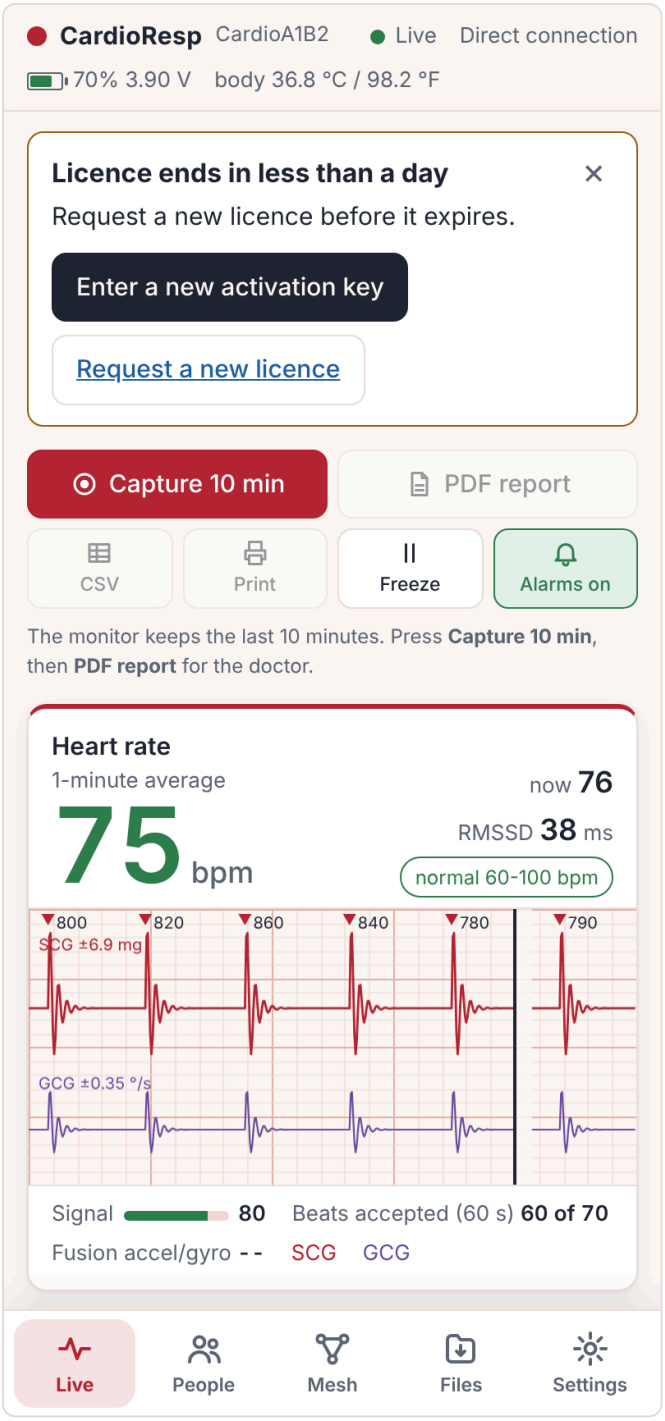
On a computer the pages are on the left; the Live page fills the rest of the screen.

Page	Use
Live	toolbar (capture, PDF report, freeze, alarms), heart and breathing, assessment, vibration and temperature, sensor placement and gravity, and the **Detailed analysis** drop-down
People	patient (age, report details, profile) and nurse on one page
Mesh	all monitors and patients on the mesh
Files	recording to the monitor's memory, PDF reports and CSV downloads of the recordings
Settings	theme, alarm sounds, calibration, power and heat, network, system, licence, firmware and About

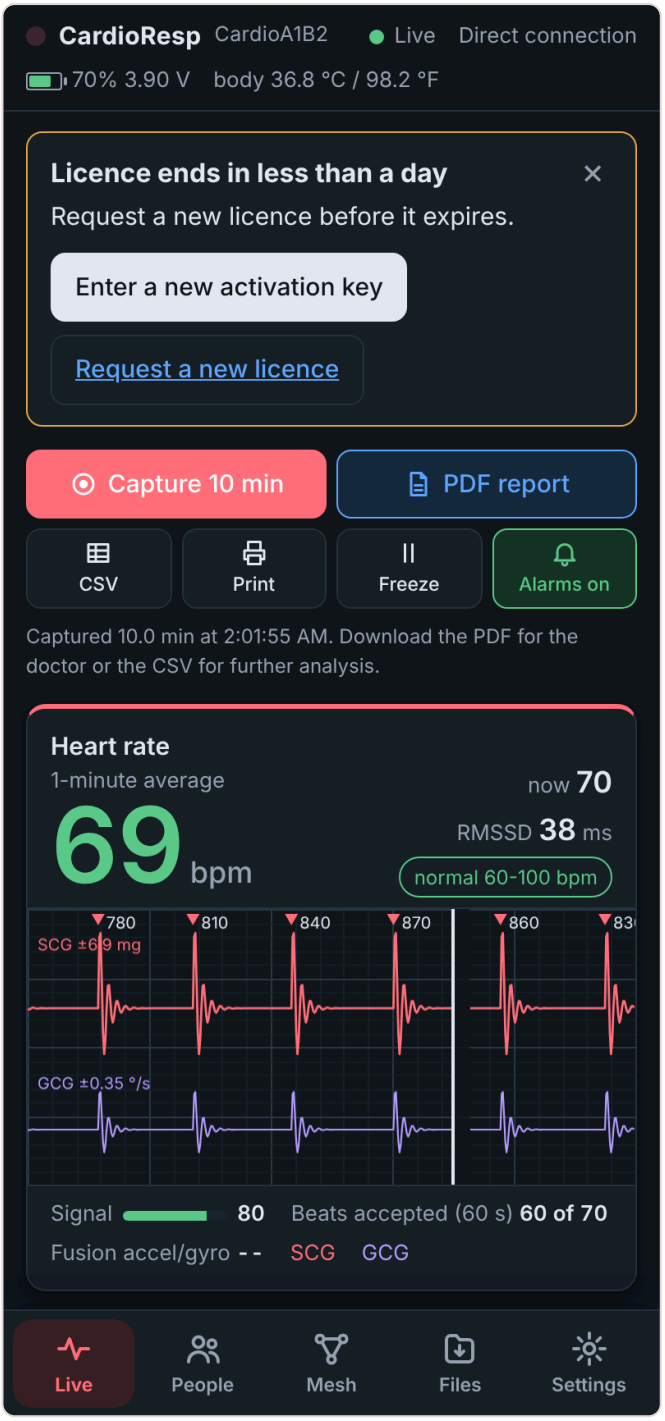
The page address ends in `#live`, `#people` and so on (the old addresses `#patient`, `#nurse`, `#analysis` and `#about` still work and jump to the right place), so you can bookmark any page. Only the visible page is drawn, which saves phone battery.

6. Live page

From top to bottom the Live page shows: the **toolbar**, the captured report (after a capture), the heart rate card, the breathing rate card, the assessment card, the vibration and temperature card, the **Sensor placement and gravity** card (chapter 7) and the **Detailed analysis** drop-down (chapter 8). Screenshots use demonstration data with a simulated heartbeat.



Live page on a phone (light theme).



The same page in the dark theme.

Toolbar

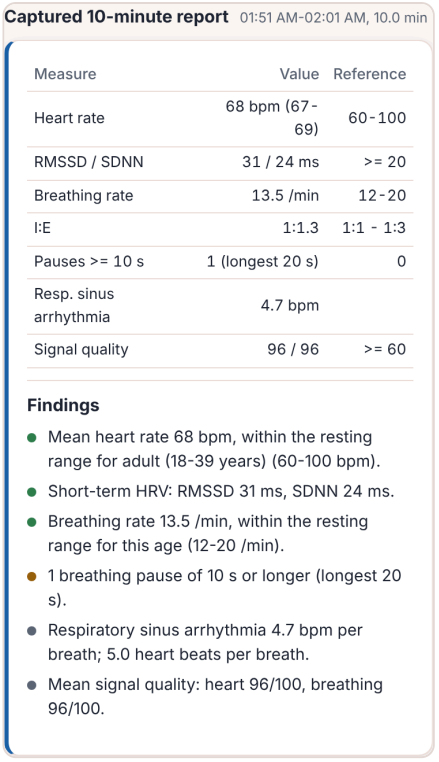


Live toolbar. PDF report, CSV and Print become active after a capture.

Button	What it does
Capture 10 min	fetches the last 10 minutes kept by the monitor and builds a report (needs a licence). The report summary appears just below the toolbar.
PDF report	downloads the 3-page PDF of the captured 10 minutes: patient details (from the People page), summary, interpretation, trends, breathing waveform, tachogram, Poincaré plot, spectra and events
CSV	every value of the capture: summary, 1 Hz trend, each beat, each breath, events, waveforms
Share / Print	shares the PDF (phones that support it) or prints the captured report
Freeze	holds the traces on screen for a closer look (amber while frozen); the monitor keeps measuring
Alarms on / off	switches all alarm sounds on or off for this page (amber when off). Opening the page again always switches them back on.



After **Capture 10 min**.



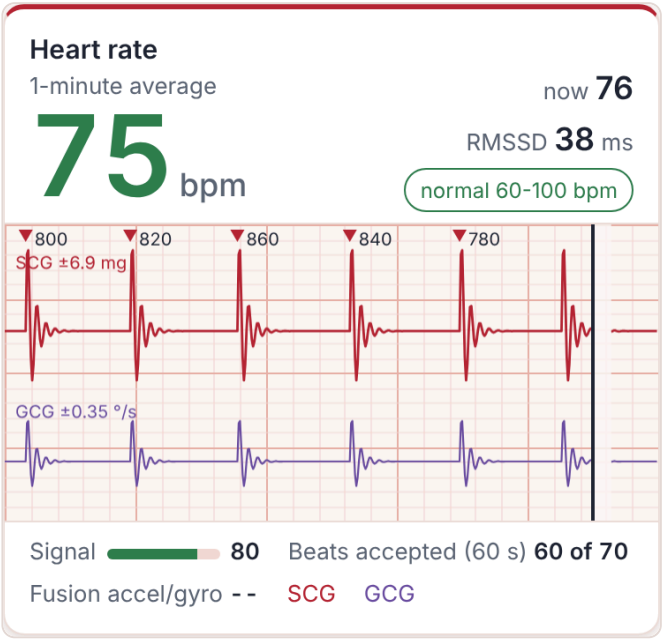
Captured 10-minute report summary.

Typical use: enter the age and the report details on the People page, then press **Capture 10 min** and **PDF report**.

> **Note.** If the page was opened inside the phone's Wi-Fi sign-in window, downloads may be blocked. Open `http://192.168.4.1` in the normal browser.

Heart rate card

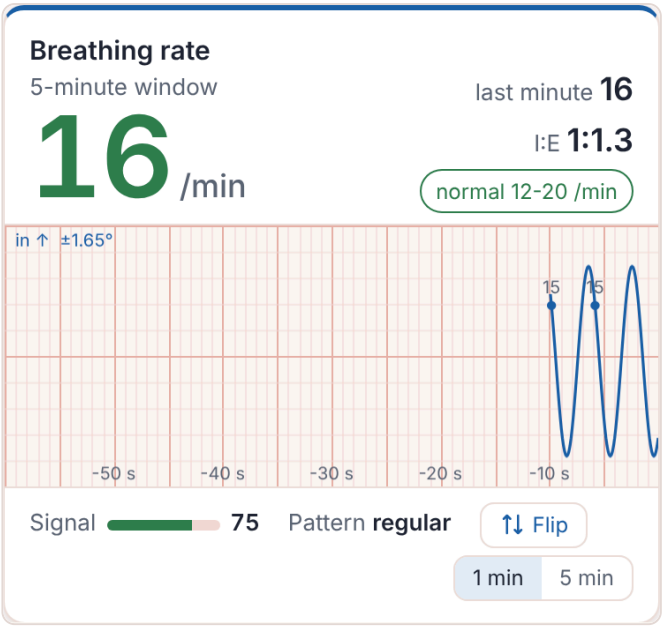
- **Big number:** heart rate averaged over the last minute, in beats per minute (bpm).
- **now:** instantaneous rate.
- **RMSSD:** beat-to-beat variability in ms (available after about 1 minute).
- **Range chip:** "normal 60-100 bpm" (green) or "above/below ..." (red). The limits depend on the age you entered.
- **Trace:** the last 5 seconds of chest vibration on ECG-style paper. SCG (red) is the vibration, GCG (violet) is the rotation. Triangles mark detected beats with the interval in ms.
- **Signal:** signal quality 0-100 (green ≥ 60, amber 30-59, red < 30).
- **Beats accepted (60 s):** beats used out of beats found.



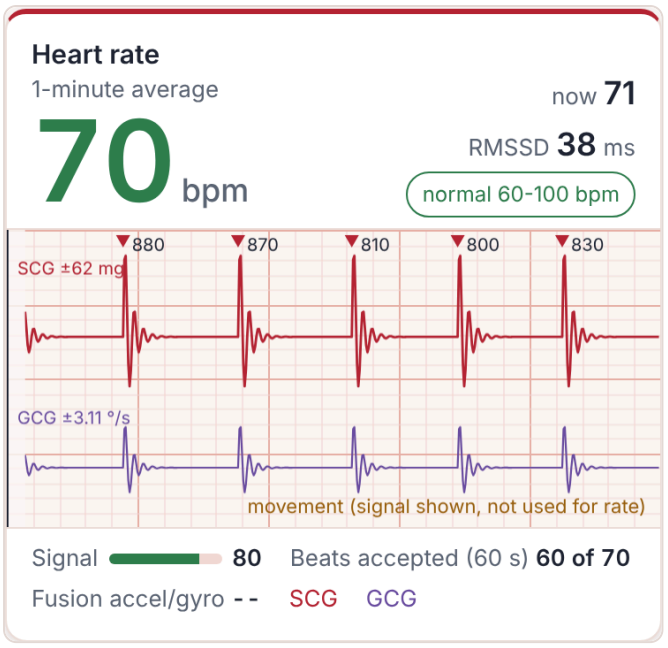
- **Fusion accel/gyro:** how much each sensor contributes.

Breathing rate card

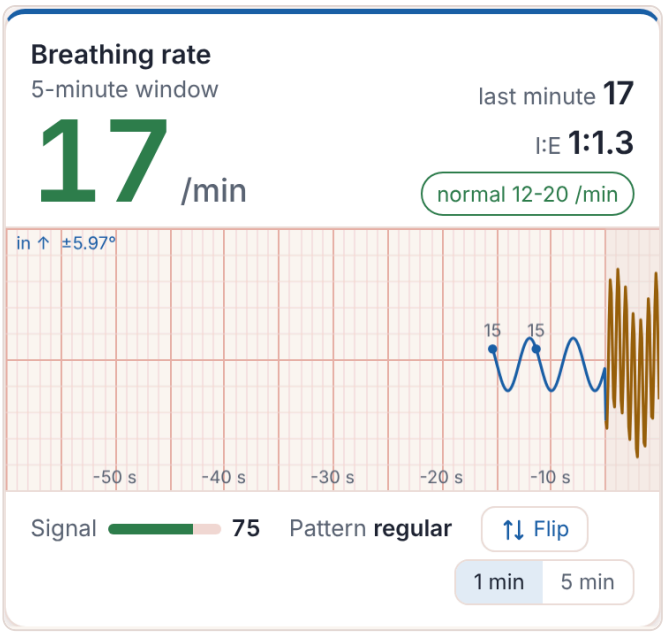
- **Big number:** breaths per minute over the last 5 minutes.
- **last minute:** the faster-reacting 1-minute rate.
- **I:E:** ratio of inhale to exhale time.
- **Trace:** chest tilt, inhale upward. Dots mark breaths with their rate. Amber = movement (shown, but left out of the rate). Pink = a breathing pause in progress.
- **1 min / 5 min** buttons change the trace window.
- **Flip** turns the breathing trace upside down if breathing in is drawn downwards (the trace normally corrects itself within about 10 breaths).
- **Pattern:** regular, irregular, shallow, with pauses, sighs, periodic and so on.



> **Traces are never hidden.** The heart (SCG, GCG) and breathing traces are always drawn, including while the patient moves, coughs or the contact is poor. Movement is only **marked**: the breathing trace turns amber and the background is shaded, and the heart trace carries the label "movement". Beats found during movement are left out of the heart-rate number, and the vertical scale grows so a large movement is shown instead of being cut off. If a rate cannot be measured reliably the big number shows **--** and is greyed; the monitor never invents a value from noise.



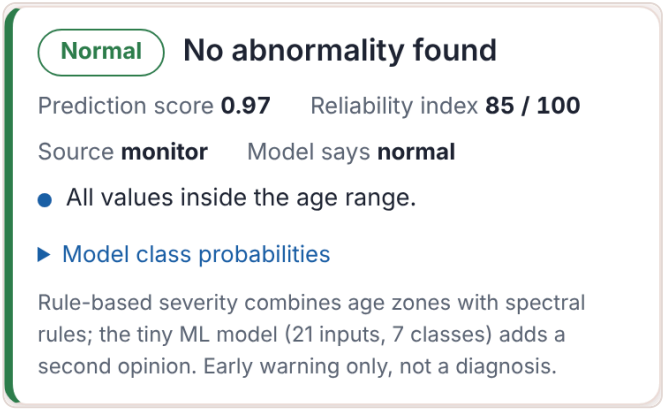
Heart trace during movement: still drawn, larger scale.



Breathing trace during movement: amber and shaded, not blanked.

Assessment card

This card is the written result of the rules and the machine-learning model.

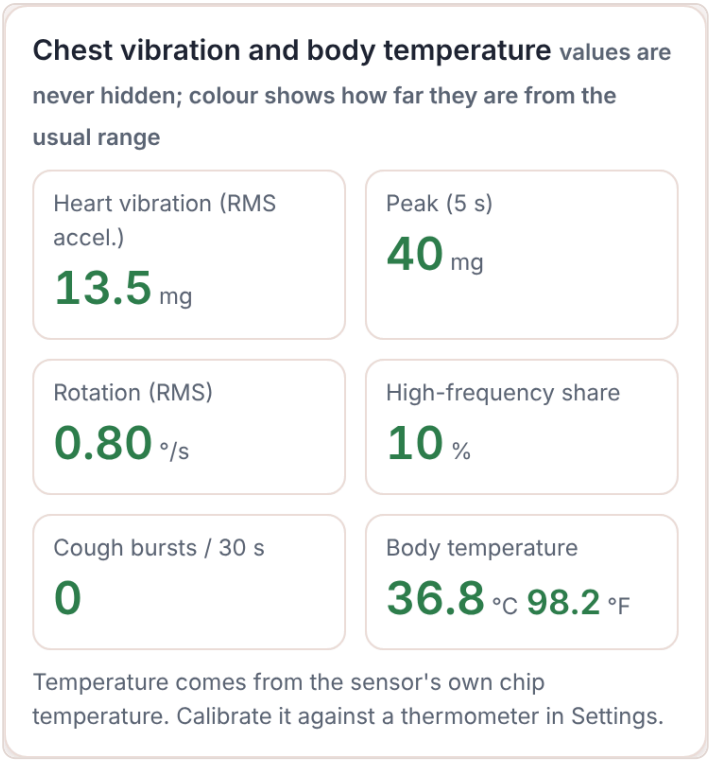


Normal result

Item	Meaning
Badge	severity: Normal, Watch, Warning, Alert (chapter 10)
Headline	one-line summary
Suspected condition	for example "Suspected tachycardia (132 bpm)"
Prediction score	model confidence in its top class, 0 to 1
Reliability index	0 to 100: how much to trust this result (signal quality, movement, agreement between heart-rate estimators)
Source	**monitor** (computed on the monitor itself) or **this page** (computed in your browser when the monitor result is missing)
Model says	the model's class; the page's class is shown if it differs
Evidence list	bullets: red = alarming, amber = warning, blue = information
Missing inputs	what could not be assessed yet (for example "needs 1 minute of still recording")

Open **Model class probabilities** to see the model's probability for each of the seven classes: normal, tachycardia, bradycardia, irregular rhythm, fast/laboured breathing, apnoea or slow breathing, coughing.

Chest vibration and body temperature card

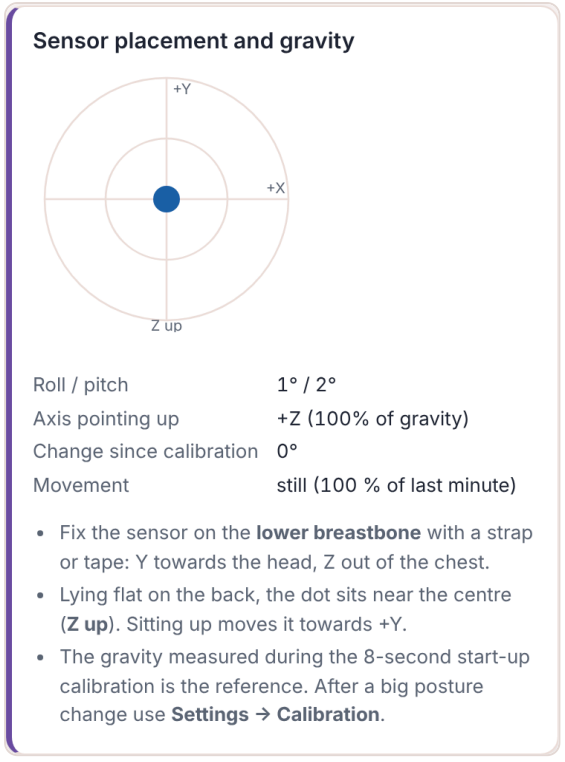


Tile	Meaning
Heart vibration (RMS accel.)	strength of chest vibration in mg
Peak (5 s)	largest vibration in the last 5 seconds
Rotation (RMS)	chest rotation rate in °/s
High-frequency share	% of vibration above 25 Hz; rises with coughing and rattling
Cough bursts / 30 s	count of cough-like bursts
Body temperature	°C and °F

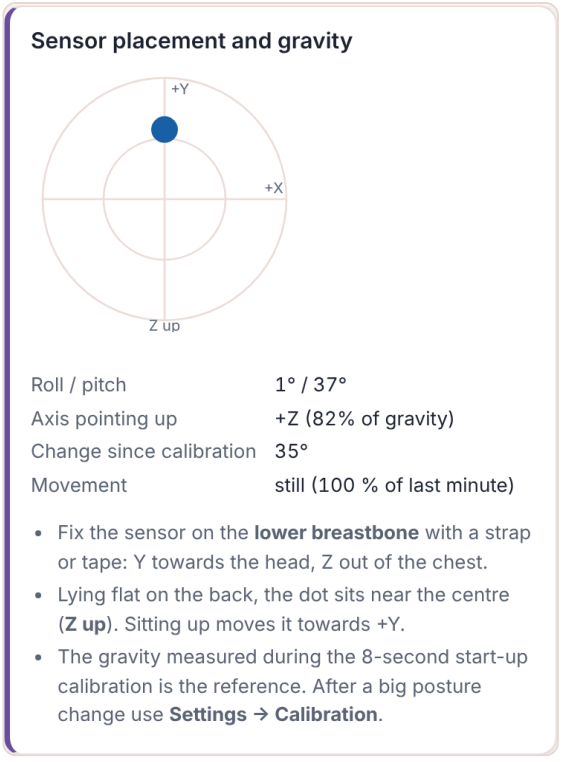
Values are never hidden. When a value leaves the usual range it is still shown, and its colour changes (next chapter).

7. Sensor placement and gravity

This card shows how the sensor sits with respect to **gravity**, the pull of the earth that always points straight down. The accelerometer measures it in the sensor's own X, Y and Z axes.



Lying flat on the back: the dot is in the centre (Z up).



Sitting up about 35°: the dot moves towards +Y.

Item	Meaning
Circle with a dot	a "bubble level" seen from above the sensor. The dot shows where **up** is in the sensor's X/Y plane; blue when Z points up, red when the sensor is upside down. The inner ring is about 30° of tilt.
Roll / pitch	tilt about the sensor's X and Y axes in degrees
Axis pointing up	which sensor axis points most upwards, and how much of gravity it carries
Change since calibration	how far the posture has turned since the 8-second start-up calibration (the gravity reference)
Movement	moving now, or the share of the last minute that was still

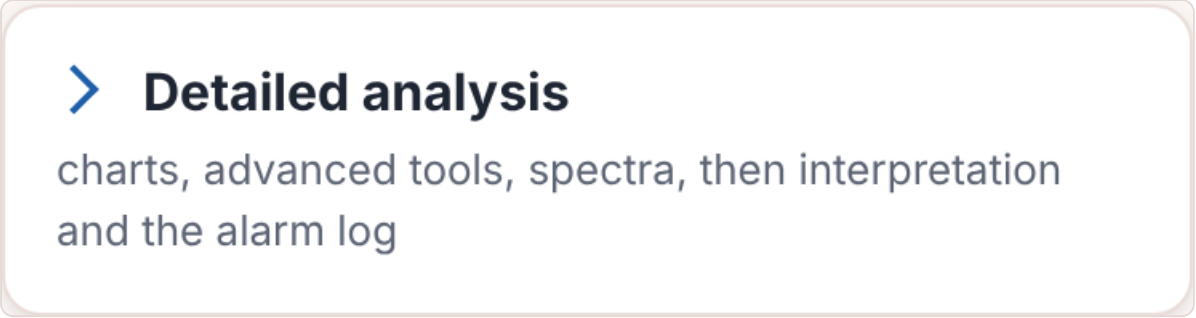
Placing the sensor with the card

1. Fix the sensor on the lower breastbone with a strap or tape: **Y towards the head, Z out of the chest**.
2. With the person lying flat on the back, the dot should sit near the centre and the axis pointing up should be +Z.
3. Keep still for the start-up calibration. "Change since calibration" should then read about 0°.
4. After a large posture change (for example lying to sitting) recalibrate in **Settings → Calibration** while the person keeps still for 8 seconds.

Breathing itself rocks the chest by only about 0.3 to 2°, which is why a firm, stable mounting matters. The card needs a valid licence (it is hidden in basic mode).

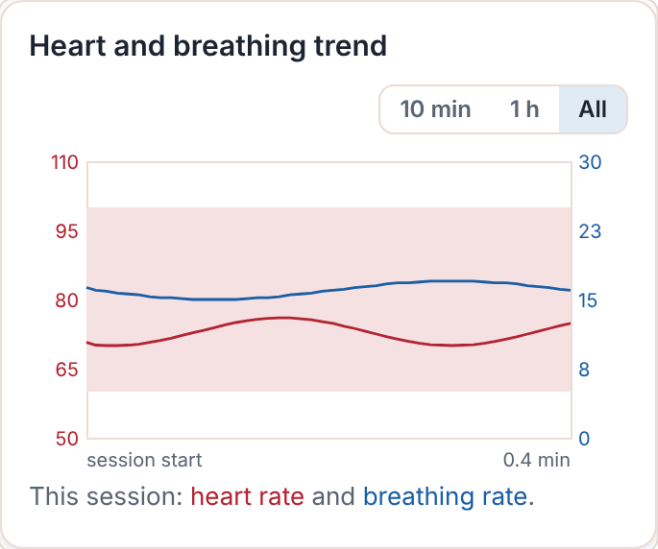
8. Detailed analysis and tools

To keep the Live page simple, the detail behind the numbers sits in the **Detailed analysis** drop-down at the bottom of the Live page. It is closed by default; tap it to open, tap again to close. The phone remembers whether you left it open, and the charts are only drawn while it is open, which saves battery. Inside, **all graphs come first and the written text follows below**.

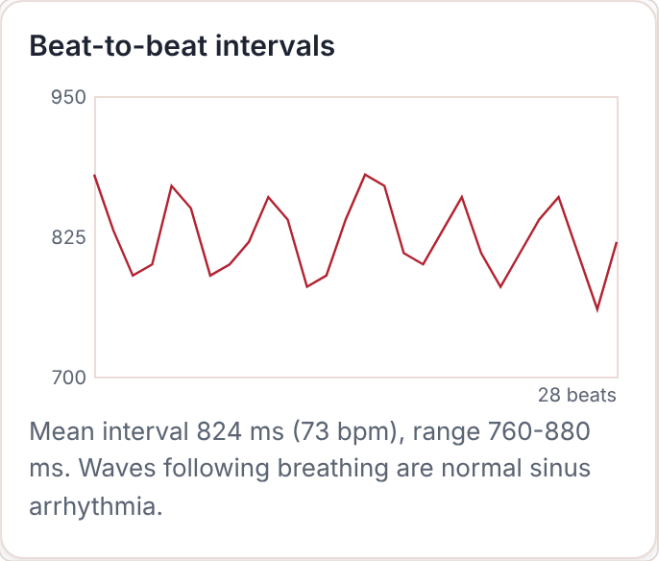


Closed drop-down.

Charts

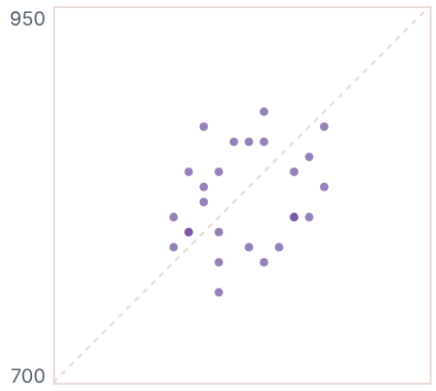


Heart (red) and breathing (blue) trend. The buttons choose the last 10 minutes, the last hour or the whole session. The shaded band is the normal heart-rate range.



Beat-to-beat intervals (tachogram). Slow waves that follow breathing are normal.

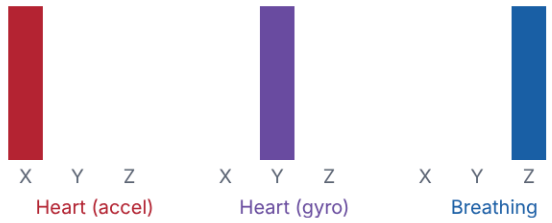
Poincaré plot



SD1 27 ms (beat-to-beat), SD2 36 ms (longer-term). A tight, elongated cloud along the diagonal is typical of a regular sinus rhythm.

Poincaré plot: each interval against the next. A tight cloud means a regular rhythm; a wide or fan-shaped cloud means irregular intervals.

Dominant signal axes (adaptive PCA)

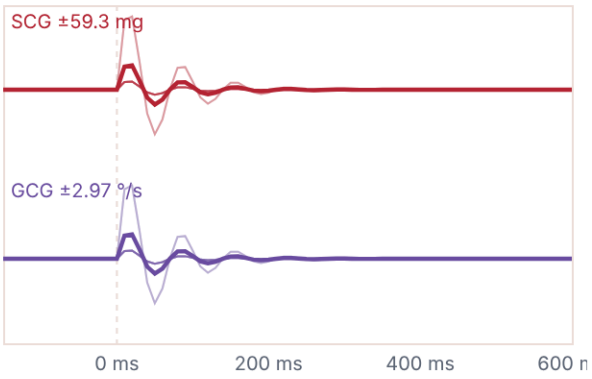


The processor finds, in any placement, the 3-D direction carrying most heart vibration (accelerometer and gyroscope) and most breathing tilt, and fuses them by signal quality.

Dominant signal axes: the directions that carry most heart vibration and most breathing tilt, found automatically for any placement.

Advanced tools

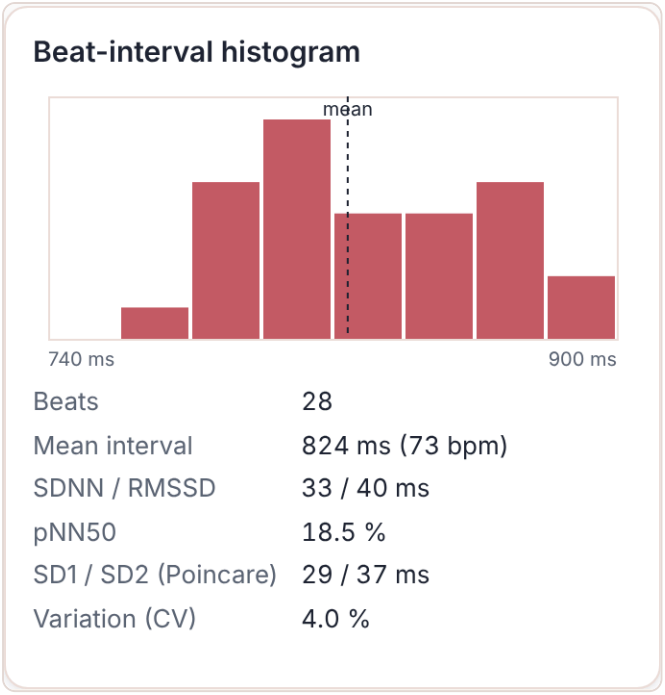
Average heartbeat



12 beats averaged. Peak-to-peak 30.3 mg; strongest deflection 20 ms after the beat marker. Beat-to-beat consistency 100 % (very similar beats, good contact).

Average heartbeat.

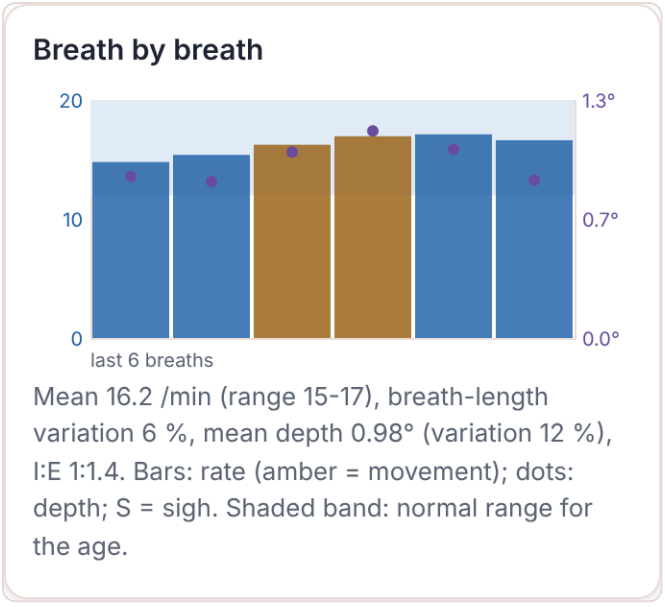
Average heartbeat. The chest vibration (red) and rotation (violet) of the recent beats are laid on top of each other, aligned on the beat marker, and averaged (thick line). Thin lines are single beats. Below the chart: the number of beats, the size of the vibration, when the strongest movement happens after the marker, and the **beat-to-beat consistency**. High consistency (above 80 %) means good sensor contact and a steady rhythm; low consistency points to movement, weak contact or an irregular rhythm.



Beat-interval histogram with variability statistics.

Beat-interval histogram. How often each beat-to-beat interval occurred (20 ms columns, dashed line = mean). A narrow single peak is a regular rhythm; a wide or split histogram means irregular intervals. Under it:

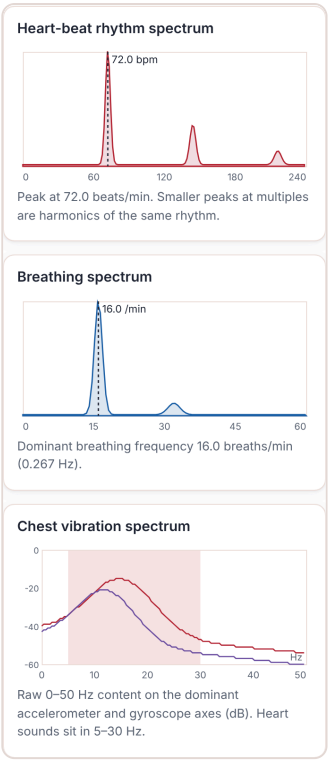
Value	Meaning
Mean interval	average time between beats (and the matching heart rate)
SDNN	overall spread of the intervals
RMSSD, pNN50	short-term, beat-to-beat variation
SD1 / SD2	width and length of the Poincaré cloud
Variation (CV)	spread as % of the mean



Breath by breath.

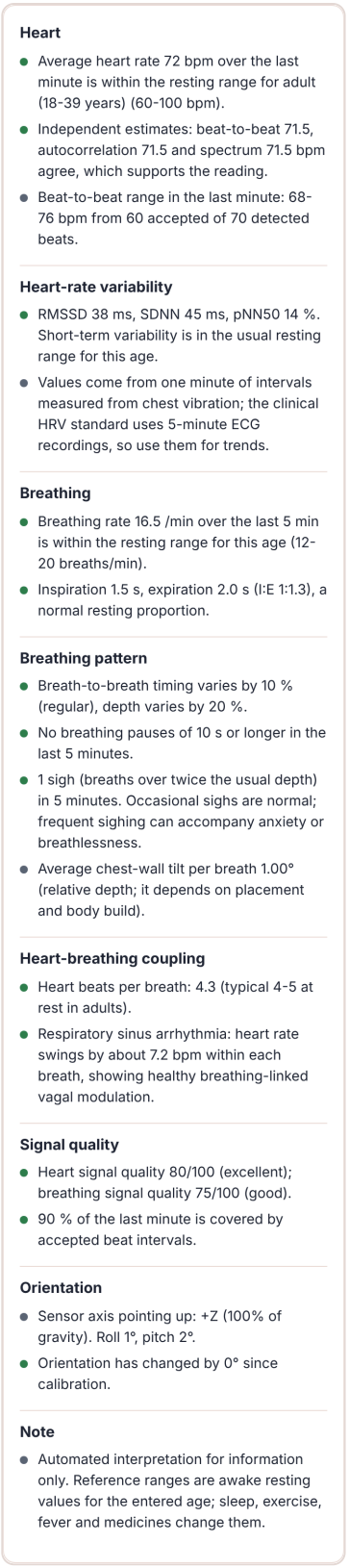
Breath by breath. Each bar is the rate of one breath (amber when the person was moving), each dot its depth; **S** marks a sigh. The shaded band is the normal breathing range for the age. The text gives the mean rate and range, how much breath length and depth vary, and the in:out (I:E) ratio. Very variable breaths, repeated sighs or a rate outside the band are worth a closer look.

Frequency analysis



Heart-beat rhythm spectrum (tallest peak = heart rate), breathing spectrum and 0-50 Hz chest vibration spectrum (heart sounds sit in 5-30 Hz).

Interpretation (text)



Heart

- Average heart rate 72 bpm over the last minute is within the resting range for adult (18-39 years) (60-100 bpm).
- Independent estimates: beat-to-beat 71.5, autocorrelation 71.5 and spectrum 71.5 bpm agree, which supports the reading.
- Beat-to-beat range in the last minute: 68-76 bpm from 60 accepted of 70 detected beats.

Heart-rate variability

- RMSSD 38 ms, SDNN 45 ms, pNN50 14 %. Short-term variability is in the usual resting range for this age.
- Values come from one minute of intervals measured from chest vibration; the clinical HRV standard uses 5-minute ECG recordings, so use them for trends.

Breathing

- Breathing rate 16.5 /min over the last 5 min is within the resting range for this age (12-20 breaths/min).
- Inspiration 1.5 s, expiration 2.0 s (I:E 1:1.3), a normal resting proportion.

Breathing pattern

- Breath-to-breath timing varies by 10 % (regular), depth varies by 20 %.
- No breathing pauses of 10 s or longer in the last 5 minutes.
- 1 sigh (breaths over twice the usual depth) in 5 minutes. Occasional sighs are normal; frequent sighing can accompany anxiety or breathlessness.
- Average chest-wall tilt per breath 1.00° (relative depth; it depends on placement and body build).

Heart-breathing coupling

- Heart beats per breath: 4.3 (typical 4-5 at rest in adults).
- Respiratory sinus arrhythmia: heart rate swings by about 7.2 bpm within each breath, showing healthy breathing-linked vagal modulation.

Signal quality

- Heart signal quality 80/100 (excellent); breathing signal quality 75/100 (good).
- 90 % of the last minute is covered by accepted beat intervals.

Orientation

- Sensor axis pointing up: +Z (100% of gravity). Roll 1°, pitch 2°.
- Orientation has changed by 0° since calibration.

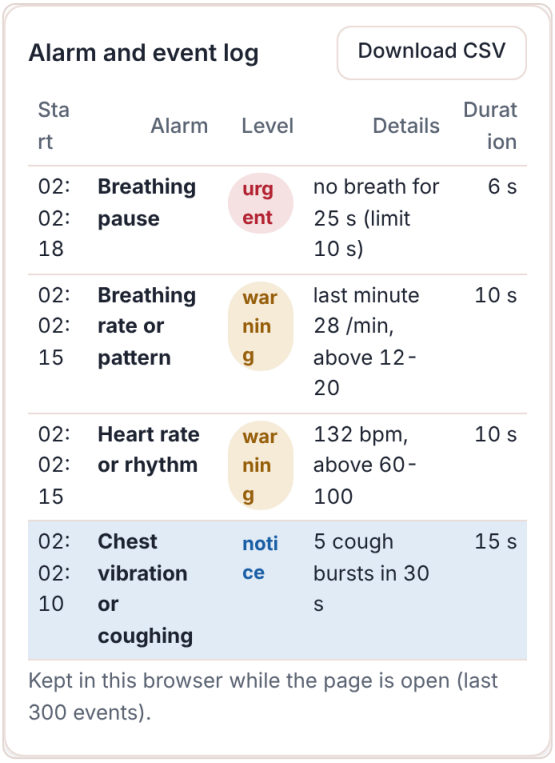
Note

- Automated interpretation for information only. Reference ranges are awake resting values for the entered age; sleep, exercise, fever and medicines change them.

Written interpretation, updated every 4 seconds.

Heart, heart-rate variability, breathing, breathing pattern, heart-breathing coupling (respiratory sinus arrhythmia), signal quality and orientation, each with findings marked green, amber or red.

Alarm and event log



Alarm and event log.

Every alarm since the page was opened: start time, alarm, level (urgent, warning, notice), details and how long it lasted ("active" while it still runs). **Download CSV** saves the log as a spreadsheet file. The log is kept in this browser while the page is open; it starts again when the page is reopened.

Recent breaths lists the time, cycle length, in and out times, I:E and depth of the latest breaths.

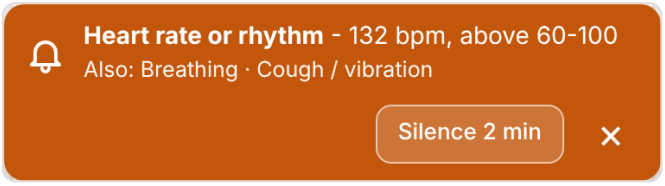
> **Tip.** If the breathing trace looks upside down (I:E above 1) it corrects itself within about 10 breaths, or press **Flip** in the breathing card.

9. Alarm sounds

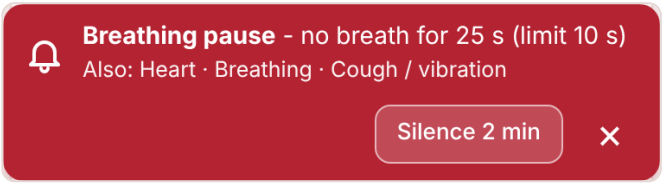
The page plays an alarm sound when a measurement needs attention. **Each kind of alarm has its own sound**, so you can tell from the sound alone what is wrong, and a coloured **alarm bar** appears above the navigation bar on every page.

Alarm	Starts when	Priority	Sound
Heart signal lost	the assessment reports that the heartbeat vibration was lost while the person lies still (needs a licence)	urgent	fast high burst (3 + 2 beeps)
Breathing pause	no breath for longer than the pause limit of the age group (adult 10 s, child 15 s, infant 20 s); urgent from 20 s. Works without a licence.	warning / urgent	high-low siren
Heart rate or rhythm	heart rate clearly outside the age range (orange or red zone) for 10 s with good signal; urgent when far outside; or an irregular-rhythm warning	warning / urgent	three equal beeps
Breathing rate or pattern	last-minute breathing rate clearly outside the range for 10 s; urgent when far outside; laboured breathing or repeated pauses	warning / urgent	three falling tones
Body temperature	38.0 °C or more, or below 34.5 °C, for 20 s; warning from 39.5 °C. Only after the temperature has been calibrated (Settings).	notice / warning	two slow tones
Chest vibration or coughing	repeated cough bursts, or strong chest vibration while the person is still, for 5 s (needs a licence)	notice	two rising chirps
Sensor lost	the sensor cable is disconnected for 3 s	warning	two low buzzes
No data from the monitor	nothing received for 15 s (out of range, monitor off or restarted); urgent if an urgent alarm was active	warning / urgent	three low buzzes

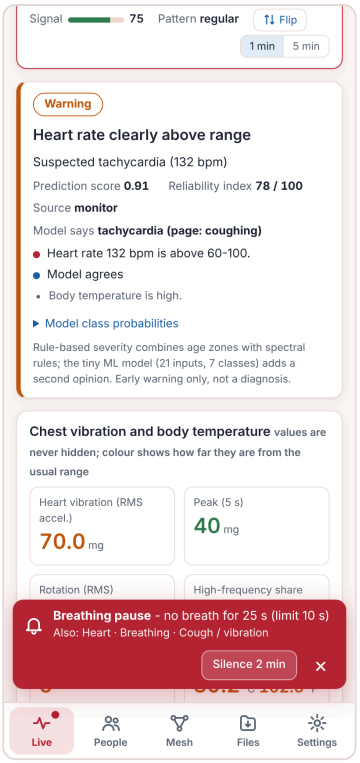
Urgent alarms repeat every 4 seconds and also vibrate the phone; **warnings** repeat every 10 seconds; **notices** every 20 seconds. Only the most urgent alarm sounds; all active alarms are listed on the bar. A condition has to be present for most of its hold time before the alarm starts, so one odd reading never starts an alarm, and an alarm ends 3 seconds after its condition is gone.



Warning (orange bar).



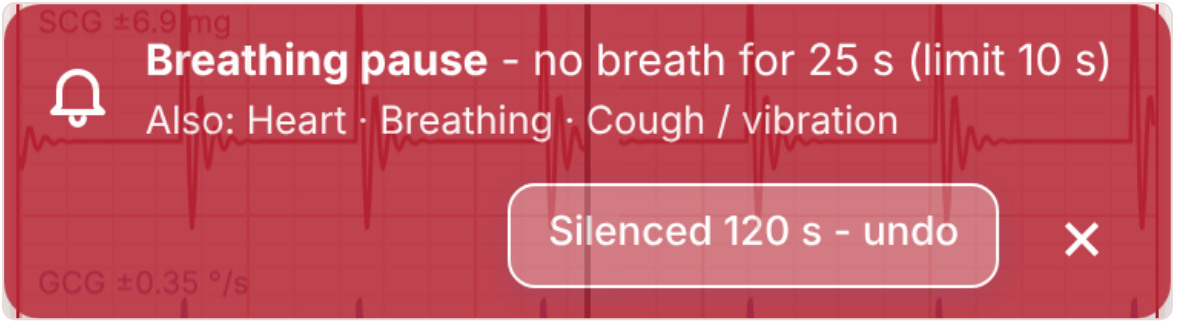
Urgent (red, pulsing bar). "Also" lists the other active alarms.



An urgent breathing-pause alarm on the Live page; the card of the affected vital sign gets a red outline and Live shows a red dot.

Silencing

- Press **Silence 2 min** on the alarm bar to mute the alarms that are active now. The bar stays visible.
- A **new** alarm, an alarm that **comes back** after it had ended, or an alarm that gets **worse** (warning to urgent) still sounds during the silence. Press **Silence 2 min** again to silence it too.
- While nothing new is sounding, the button reads **Silenced ... s - undo**; press it to end the silence at once.
- The **×** on the bar does the same as Silence and also hides the bar. It comes back by itself when a new or worse alarm starts.
- Every alarm is recorded in the **Alarm and event log** under Detailed analysis (chapter 8).



Silenced alarm bar.

Making sure you hear the alarms

- Browsers only allow sound after the page has been **tapped once**. Until then the alarm bar says "Tap the screen once to allow the alarm sound".
- Keep this page **open with the screen on**. A locked phone or a page in the background may stop the sound.
- Check the phone's media volume and the volume slider in **Settings → Alarm sounds**; use the **Test** buttons to hear each sound.
- The toolbar's **Alarms on/off** button mutes everything for this page only; opening the page again switches the alarms back on.

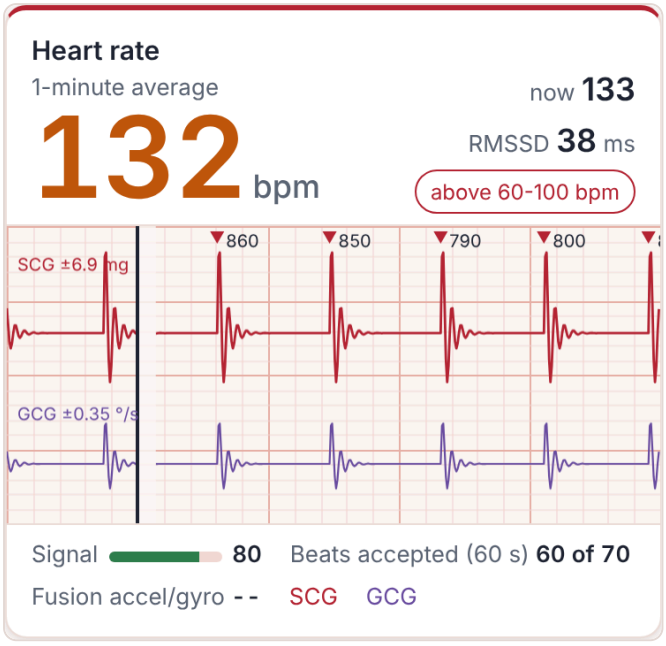
> **Not a certified alarm system.** The alarms are an early-warning aid on a phone or computer. They depend on the browser, the screen and the Wi-Fi link. Do not rely on them as the only safety monitoring.

10. Colours, severity and what to do

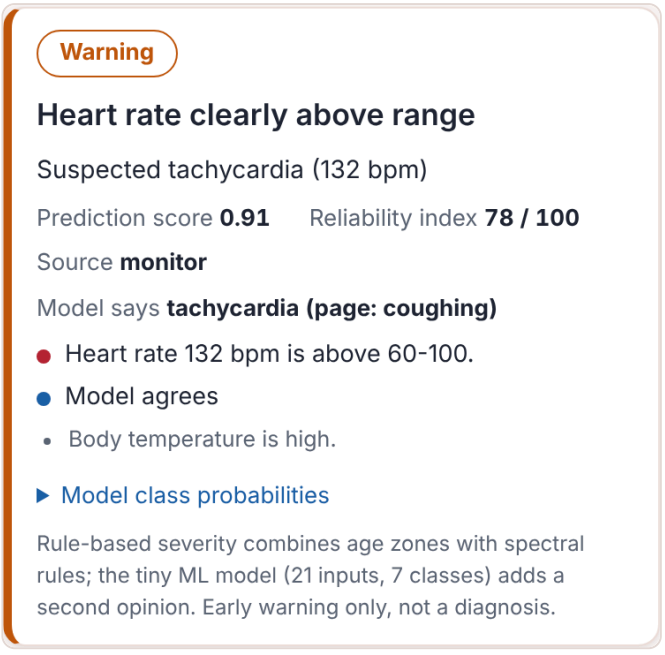
Colour scale

Colour	Zone	Meaning for heart rate / breathing rate
Green	normal	inside the age range
Amber	watch	up to 20 % above or about 17 % below the range
Orange	warning	20-50 % above, or far below the range
Red	alert	more than 50 % above or far below the range

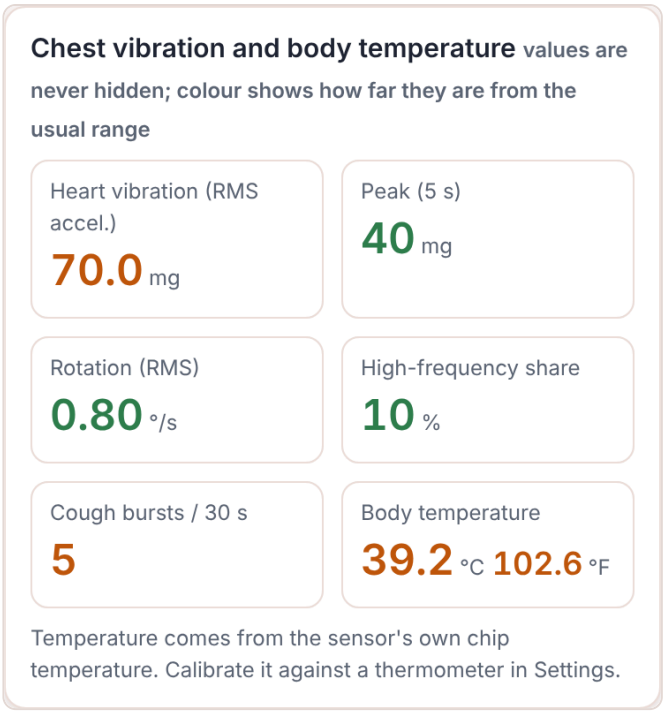
The same four colours are used for the vibration tiles and for temperature: amber from 37.6 °C (99.7 °F), orange from 38.0 °C (100.4 °F), red from 39.5 °C (103.1 °F) or below 34.5 °C (94.1 °F).



Abnormal state: heart rate 132 bpm. The number stays visible and changes colour.



Warning result with evidence



Coloured vibration and temperature tiles

Severity levels

Badge	Meaning	Suggested action
Normal	nothing outside the age-referenced limits	continue monitoring
Watch	mild deviation or the model leans towards a condition below the alert level	check sensor contact, look at the patient, re-check in a few minutes
Warning	clear deviation (for example heart rate 20-50 % above range, repeated pauses, worsening trend, cough bursts with raised temperature)	go to the patient, measure with a reference method, inform the responsible clinician
Alert	severe deviation or a dangerous pattern (very high or low rates, long breathing pause, signal pattern consistent with loss of pulse)	go to the patient immediately; check responsiveness and breathing; follow emergency procedures

Patterns the monitor looks for

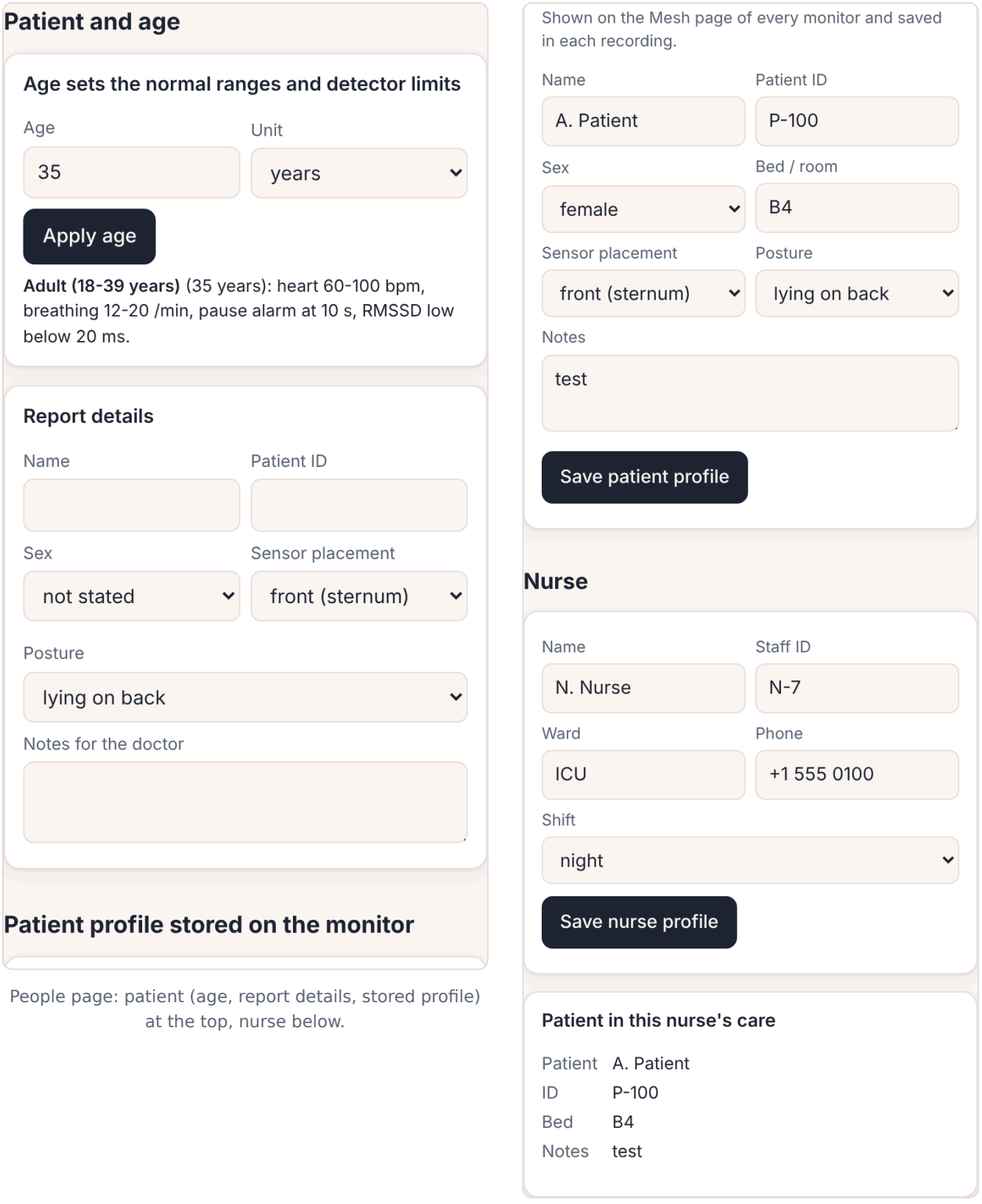
- **Tachycardia / bradycardia:** heart rate outside the age range.
- **Irregular rhythm:** high beat-to-beat variability (RMSSD, pNN50), spread of the rate and disagreement between estimators.
- **Fast or laboured breathing, slow breathing:** breathing rate and depth outside range, I:E ratio.
- **Apnoea:** breathing pause longer than the age threshold (adult 10 s, child 15 s, infant 20 s).
- **Possible cardiac arrest pattern:** a previously good heart signal disappearing while the patient is not moving, together with breathing change. This can also be a loose sensor: check the patient first, then the sensor.
- **Early heart-attack hint:** sustained abnormal rate or rhythm with a worsening trend. Chest vibration cannot diagnose a heart attack.

- **Coughing / possible respiratory infection:** repeated cough bursts with high-frequency vibration, raised breathing rate and temperature. It cannot identify COVID or any particular infection.

> **Always verify.** The thresholds are heuristic. Treat Warning and Alert as a prompt to examine the patient, not as a diagnosis. Movement, a loose strap or a hot sensor can create false alarms; check the **Reliability index**.

Alarm sounds: every kind of problem has its own sound, and urgent alarms repeat every 4 seconds; see chapter 9.

11. People page: patient and nurse



People page: patient (age, report details, stored profile) at the top, nurse below.

Age and normal ranges

1. Type the age and choose **years** or **months** (use months for babies).

2. Press **Apply age**.

The age selects the normal heart and breathing ranges, the detector limits, the breathing filters and the pause-alarm threshold. It is saved on the monitor. The line below shows the group, for example "Adult (18-39 years)".

Group	Heart rate (bpm)	Breathing (/min)	Pause alarm
Newborn (0-1 month)	100-205	30-60	20 s
Infant (1-12 months)	100-180	30-53	20 s
Toddler (1-2 years)	98-140	22-37	15 s
Preschool (3-5 years)	80-120	20-28	15 s
School age (6-11 years)	75-118	18-25	15 s
Adolescent (12-17 years)	60-100	12-20	10 s
Adult (18+)	60-100	12-20	10 s

> **Set the correct age first.** A wrong age moves every limit and changes the colours and alerts.

Report details (kept on the phone)

Name, patient ID, sex, sensor placement, posture and notes for the doctor. They are printed on the PDF report and are remembered in the phone's browser.

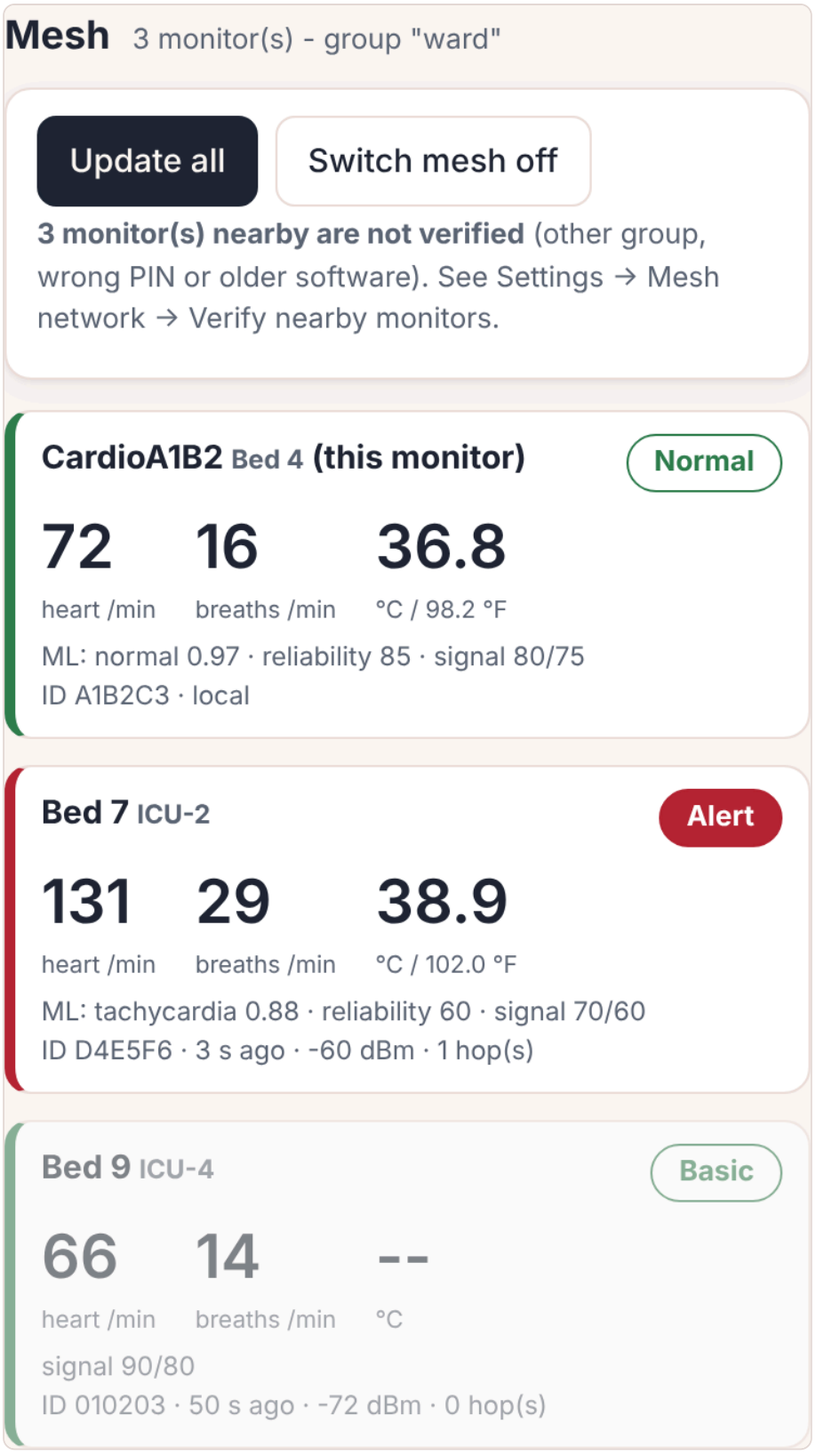
Patient profile stored on the monitor

Fill in name, ID, sex, bed/room, placement, posture and notes and press **Save patient profile**. This record is kept in the monitor's memory, shown to every monitor on the **Mesh** page and written into each recording.

Nurse

Enter name, staff ID, ward, phone and shift, then press **Save nurse profile**. The card next to it shows the patient currently assigned to this monitor, so the nurse in charge and the patient are always listed together. Patient and nurse records are separate and are saved separately.

12. Mesh page (several patients)



Mesh page with three verified monitors.

The mesh lets every monitor show the patients of the other monitors nearby over a short-range radio link. No router is needed. Only monitors with the **same group name and PIN** are shown;

their messages are signed with them, so a monitor of another ward or with the wrong PIN cannot appear on your screen.

Setting up (once per monitor)

1. Open **Settings → Mesh network** (chapter 14).
2. Enter the **group name** (for example the ward name) and the **PIN** (4 to 8 digits). Use exactly the same group name and PIN on every monitor that should share data.
3. Give the monitor a **label** such as the bed or room, tick **Mesh on** and press **Save mesh settings**.
4. Press **Verify nearby monitors**. Every monitor in range answers and is listed as verified or not.

All monitors must also use the **same Wi-Fi channel**: keep them either all on their own networks (direct mode) or all on the same router.

Reading a patient card

- Left colour bar and badge: severity (Normal, Watch, Warning, Alert). **Basic** means that monitor's licence is not active.
- Name and label (bed / room), then heart rate, breathing rate and temperature (°C / °F).
- Model class, score and reliability, signal quality for heart / breathing and a "moving" note.
- Footer: monitor ID; for others, seconds since the last update, signal strength (dBm) and hops. Cards not updated for 30 seconds are faded.
- The first card is **this monitor**. The others are sorted by severity so the most urgent patients are first.

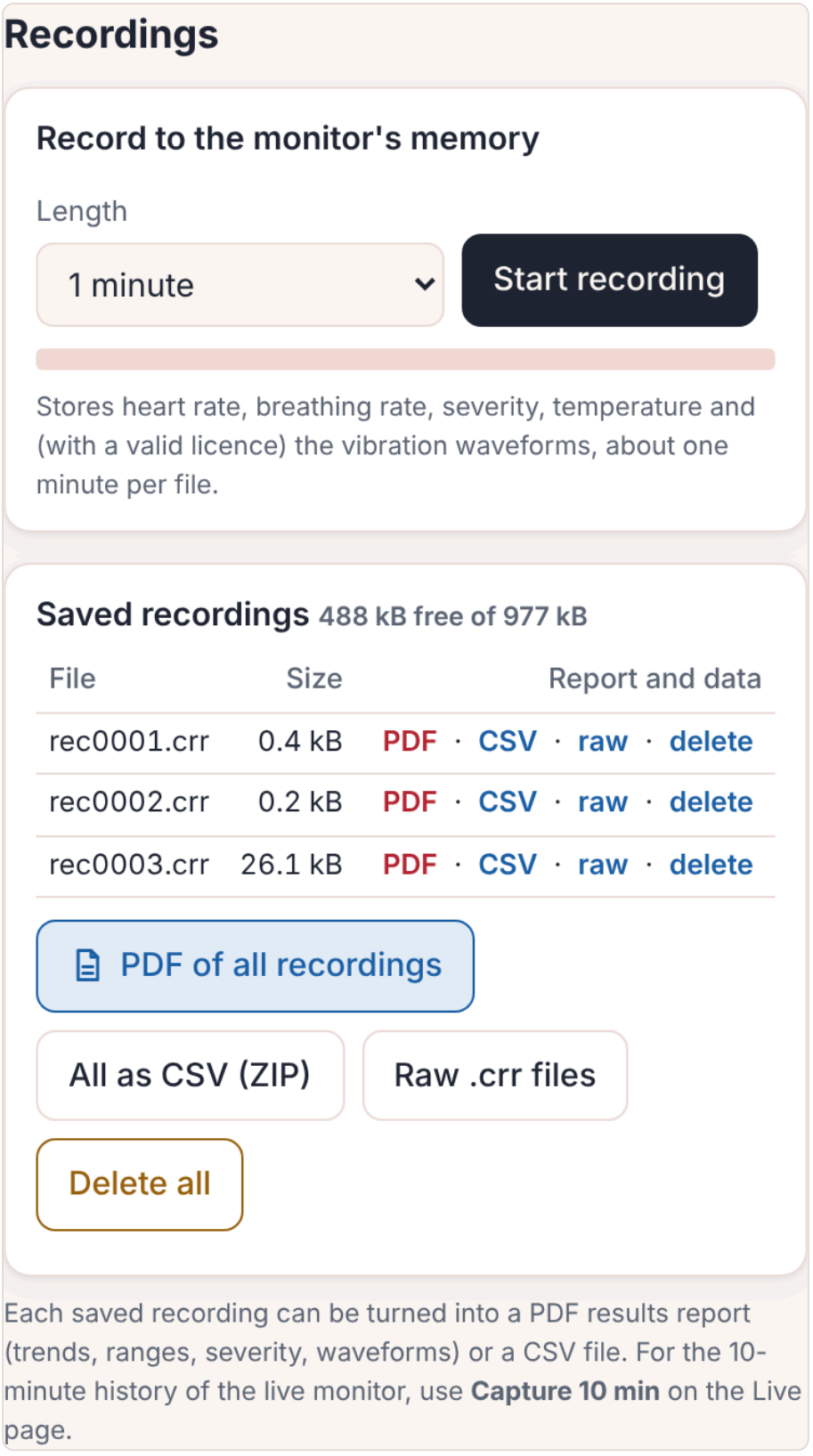
The line at the top shows the group name. If monitors nearby are heard but not verified (other group, wrong PIN or older software), the page says how many; see **Settings → Mesh network → Verify nearby monitors**.

Buttons

Button	Action
Update all	asks every monitor on the mesh to send its newest data immediately.
Switch mesh on / off	turns the mesh radio on or off for this monitor. Off reduces radio activity and heat.

> **Note.** Each monitor can list up to 16 others. With relay hops (Settings) monitors pass messages on, so monitors beyond direct radio range can still be listed.

13. Files page: recordings and PDF reports



Files page: recording and saved recordings.

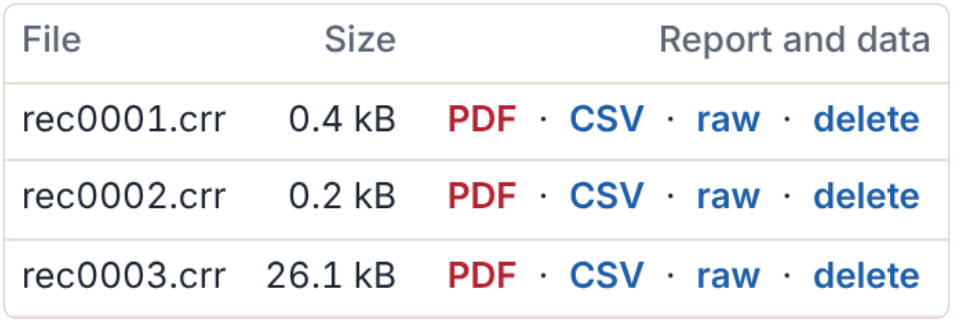
Record to the monitor's memory

1. Choose the **Length**: 30 seconds, 1 minute (default), 90 seconds or 2 minutes.
2. Press **Start recording**. The progress bar and message show "Recording rec0003.crr: 25 / 60 s". **Stop** appears while recording.
3. When it finishes the file appears under **Saved recordings**.

Each recording stores, once per second, heart rate, breathing rate, signal quality, severity, temperature and RMSSD. With an active licence it also stores the 100 Hz vibration waveform and the 10 Hz chest-tilt waveform. The patient name and ID, age, device and date are saved in the file header.

- The monitor needs the measurement running (calibration finished). If memory is short the page says so; delete old files first.
- Memory use is shown next to **Saved recordings** (for example 488 kB free of 977 kB).

Saved recordings



Each recording has PDF, CSV, raw and delete links.

Link / button	Result
PDF	a **results report** of that recording, made in your browser (see below)
CSV	converts that recording to a CSV file and downloads it
raw	downloads the original `.crr` file
delete	removes that file from the monitor
PDF of all recordings	one PDF with a report for every recording
All as CSV (ZIP)	one ZIP containing a CSV for every recording
Raw .crr files	downloads every raw file, one after another
Delete all	removes all recordings (asks to confirm)

The recording results report (PDF)

- **Header:** file name, length, device, start time and age group.
- **Patient, nurse and recording:** name, ID and age come from the recording itself; fields marked * (sex, bed, nurse, placement, posture) come from the profile stored on the monitor now.

- **Summary table:** mean, minimum and maximum of heart rate, breathing rate, RMSSD and temperature (°C and °F), the age reference range and the share of seconds outside it; signal quality and movement.
- **Assessment over time:** a colour strip with one block per second (green Normal, amber Watch, orange Warning, red Alert, grey no assessment) and the seconds in each level.
- **Findings:** plain-language statements, for example "Average heart rate 81 bpm is within the reference" or "Temperature reached 38.1 °C: possible fever".
- **Trends:** heart rate, breathing rate and temperature per second, with the reference range as a green band.
- **Waveforms** (licensed recordings): the first 10 seconds of chest vibration (SCG and GCG) and the whole breathing trace, movement included.

Layout of a downloaded CSV

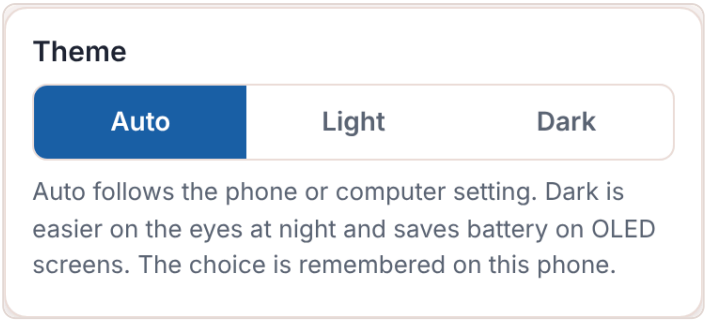
1. Header lines starting with `#`: device, patient, ID, age, start time.
2. One row per second: `second, hr_bpm, rr_per_min, signal_heart, signal_resp, severity, heart_present, motion, temp_c, temp_f, rmssd_ms`. Severity: 0 normal, 1 watch, 2 warning, 3 alert.
3. A block "vibration waveform at 100 Hz" with columns `sample, scg, gcg` (raw units).
4. A block "chest tilt at 10 Hz" in millidegrees.

These CSV files can be used directly in the training notebook (see the project README) to build a better model from your own data. For the 10-minute history of the live monitor use **Capture 10 min** on the Live page (chapter 6).

14. Settings page (including About)

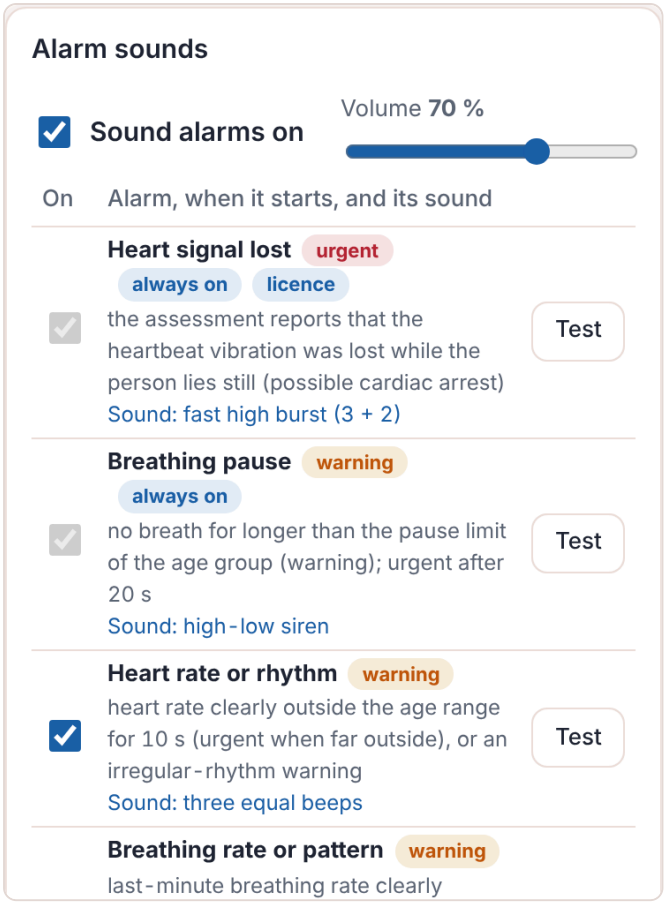
Settings is grouped into: Display and alarms, Calibration, Power / network / system, Licence and firmware, and About.

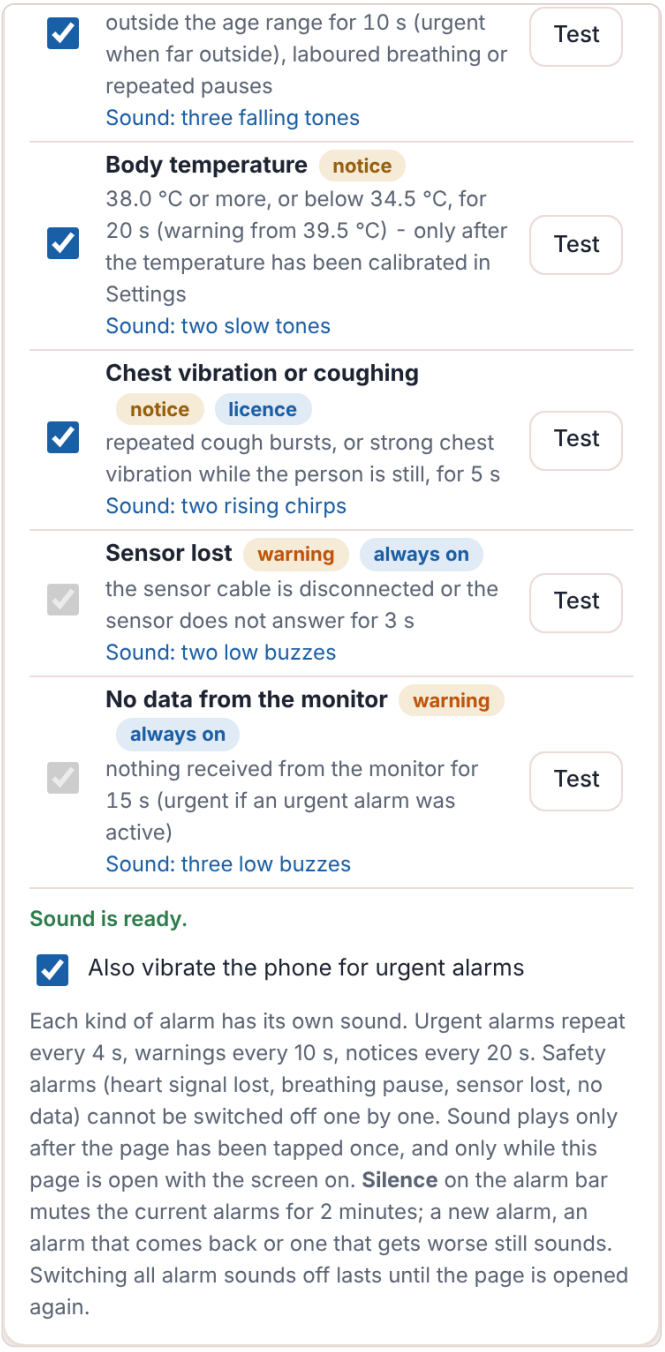
Theme



Choose **Auto** (follows the phone or computer setting), **Light** or **Dark**. Dark is easier on the eyes at night and saves battery on OLED screens. The choice is remembered on this phone.

Alarm sounds





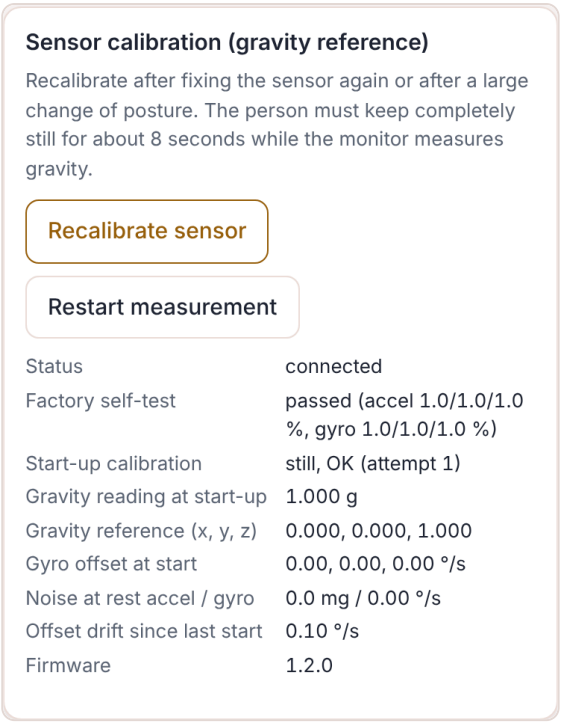
Alarm sounds card (top and bottom half).

- **Sound alarms on** and **Volume** (the slider plays a sample when released).
- A table of all alarms with when they start, their priority and their sound; **Test** plays each sound.
- Safety alarms (heart signal lost, breathing pause, sensor lost, no data) are marked **always on** and cannot be switched off one by one. The others can be switched off.
- The line under the table says whether sound is ready or still blocked by the browser.
- **Also vibrate the phone for urgent alarms** (phones that support it).

See chapter 9 for how the alarms work.

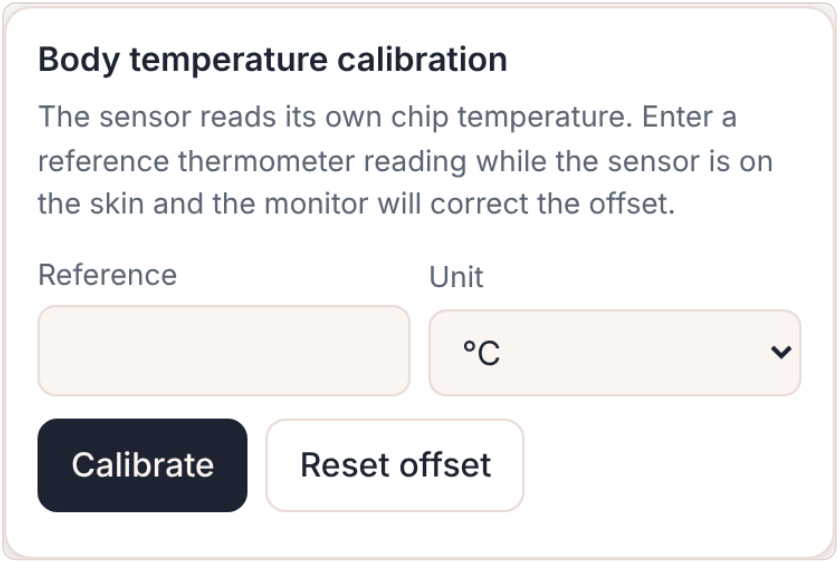
Calibration

Sensor calibration (gravity reference)



Recalibrate sensor repeats the 8-second start-up calibration: the person must keep completely still. Use it after fixing the sensor again or after a large change of posture. **Restart measurement** clears the running analysis and the 10-minute history and starts again. The card also lists the sensor details: self-test result, gravity reading, gravity reference, gyroscope offset, noise and drift.

Body temperature calibration

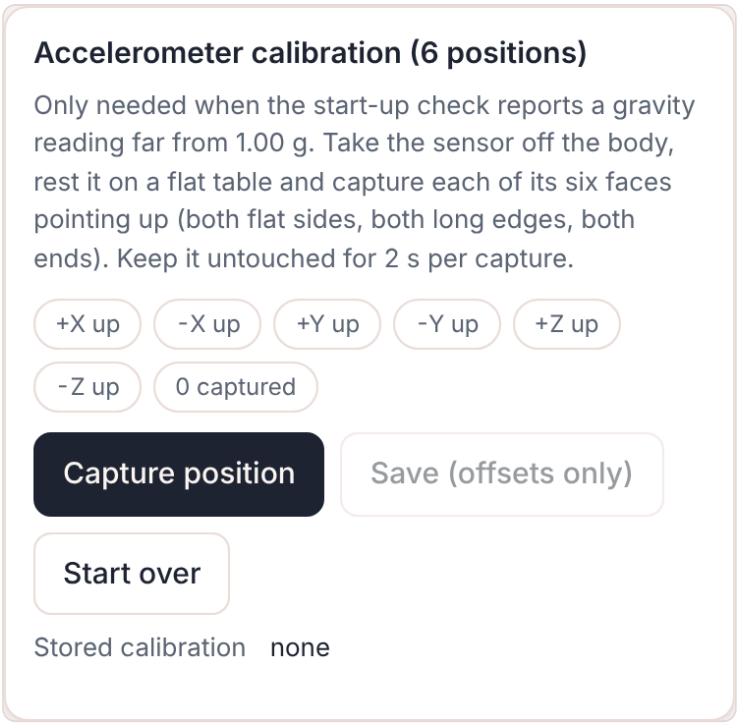


1. Put the sensor in its normal position on the body for several minutes so it warms to skin temperature.
2. Measure the person with a reference thermometer.

3. Type the reading in **Reference**, choose °C or °F and press **Calibrate**. The monitor stores the offset.
4. **Reset offset** clears it.

Valid reference range 25-45 °C (77-113 °F). The sensor measures its own temperature, so without this step the value is not a body temperature.

Accelerometer calibration (6 positions)



Only needed if the start-up check says gravity reads far from 1.00 g. Take the sensor off the body and rest it still on a flat table.

1. Place a face up (for example Z up, flat) and press **Capture position**. Do not touch it until the position is ticked.
2. Repeat for the other faces: both flat sides, both long edges and both ends. The chips +X, -X, +Y, -Y, +Z, -Z turn green.
3. Press **Save calibration** (4-5 positions save offsets only; all 6 also save gains).
4. Attach the sensor to the body and press **Recalibrate sensor**.

Start over clears the captures. **Remove stored calibration** deletes the saved correction.

Mesh network

Mesh settings

Monitors share their patients with each other over a short-range radio link; no router is needed. Only monitors with the **same group name and PIN** are accepted: every message is signed with them.

☑ **Mesh on**

Group name: ward

PIN (4 to 8 digits): •••• (unchanged) — Show

☐ No PIN (open mesh: any monitor with this group name is accepted)

Label of this monitor (bed / room): Bed 4

Send every: 2 s

Relay hops: 2

Mesh	on
Group	ward
PIN	set (4 digits)
This monitor	A1B2C3 - Bed 4
Radio channel	6

Save mesh settings

Use the same group name and PIN on every monitor that should share data. Relay hops let monitors pass on messages from monitors further away. All monitors must use the same Wi-Fi channel.

Mesh settings.

Verify nearby monitors

Asks every monitor in radio range to answer and lists them. **Verified** monitors share data with this one; the others need the same group name and PIN.

Verify nearby monitors

Monitor	Signal	Status
ICU-2 · Bed 7 ID D4E5F6	-60 dBm 3 s ago · 1 hop(s)	verified
ICU-4 · Bed 9 ID 010203	-72 dBm 50 s ago	verified
Lab bench ID 0D0E0F	-80 dBm 4 s ago	other group
ICU-6 · Bed 12 ID 0A0B0C	-66 dBm 2 s ago	wrong PIN
Old unit ID 111213	-75 dBm 6 s ago	older software

2 verified, 3 not verified. Monitors with "wrong PIN" or "other group" need the same group name and PIN as this one.

Verify nearby monitors.

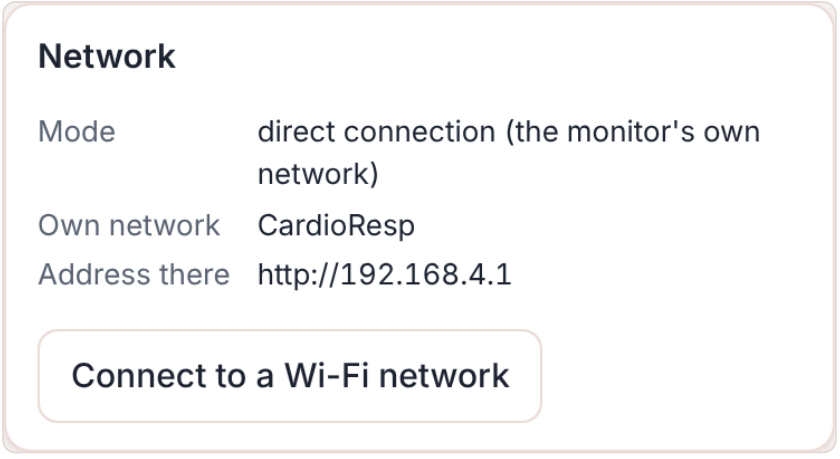
Setting	Meaning
Mesh on	switches the mesh radio on or off for this monitor
Group name	1 to 15 letters, digits, spaces, - or _ (not case sensitive). Only monitors with the same group name are listed.
PIN	4 to 8 digits. Monitors accept each other only with the same PIN. The field stays empty after saving ("unchanged"); type a new PIN only to change it. **Show** reveals what you type.
No PIN	open mesh: any monitor with the same group name is accepted. Use a PIN whenever several wards or departments are close together.
Label	bed or room shown with this monitor on the other monitors
Send every	how often this monitor sends its values (1, 2, 5 or 10 s; longer saves heat and battery)
Relay hops	how many times a message may be passed on by other monitors (0 = direct only)

Press **Save mesh settings**. Then press **Verify nearby monitors**: every monitor in range answers within a few seconds and is listed with its label, signal strength and status:

Status	Meaning and what to do
verified	same group name and PIN; it shares data with this monitor
wrong PIN	same group name but a different PIN: enter the same PIN on both monitors
other group	another group name: correct it if the monitor should belong to your group, otherwise ignore it
older software	the monitor needs a software update before it can join (Firmware update)

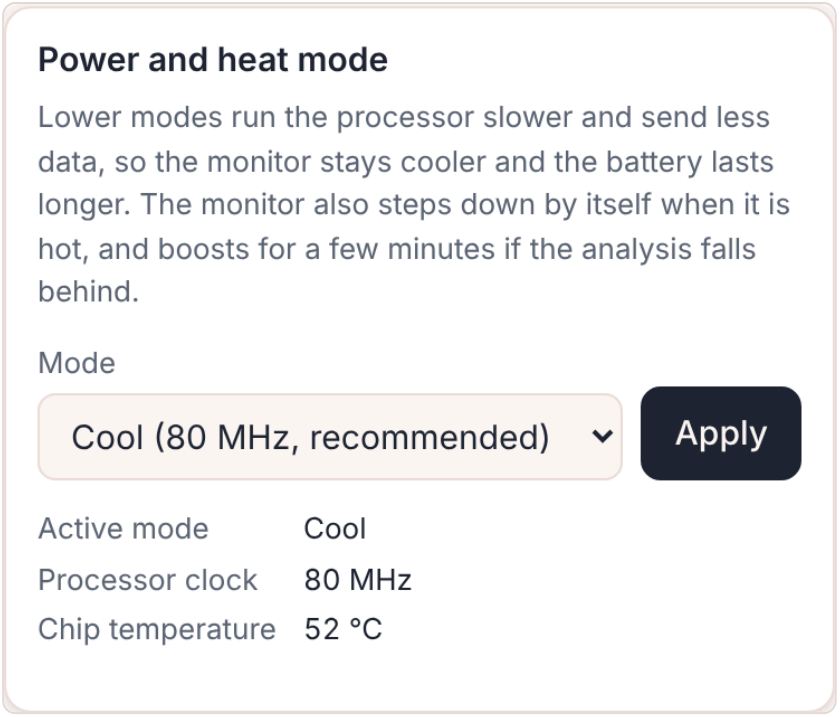
Power, network and system

Network



Shows the connection mode, own network name, the address and the saved home network. Buttons: **Connect to a Wi-Fi network** (opens the setup form of chapter 3), **Keep this address fixed** and **Forget home Wi-Fi** (returns to direct mode).

Power and heat mode

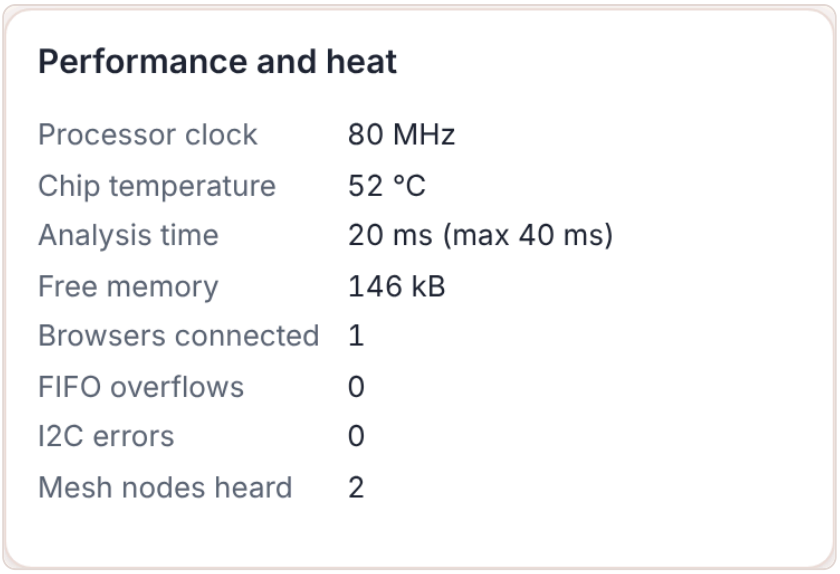


Mode	Processor	Updates	Use
Performance	160 MHz	smoothest traces (waveform 10 per second)	short bedside checks on mains power
Cool (default)	80 MHz	waveform 5 per second	normal use; clearly cooler
Eco	80 MHz	waveform 2.5 per second, status every 2 s	long recordings, battery, enclosed case

Press **Apply**; the mode is saved and also applied to the Wi-Fi radio power. The card shows the active mode, the processor clock and the chip temperature. If the chip gets hot the monitor slows down by itself, and in Cool or Eco it briefly boosts to 160 MHz (for up to 5 minutes) if the analysis falls behind. Recording, heart rate and breathing results are the same in all three modes.

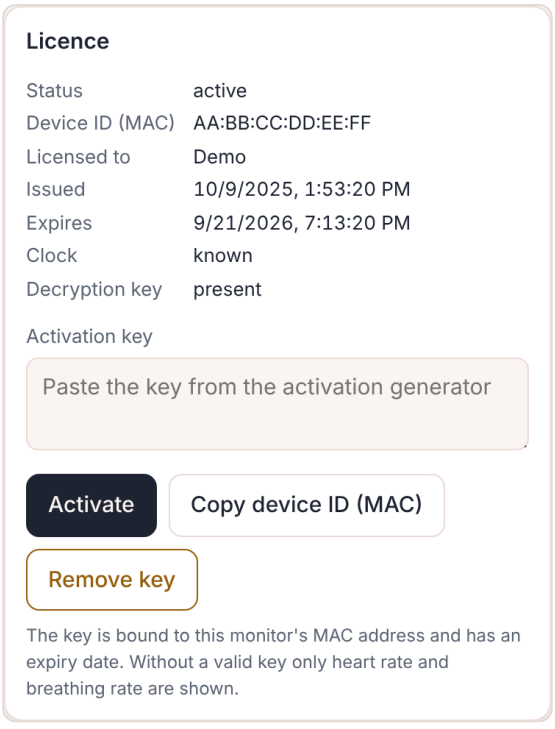
System and performance cards

- **System:** firmware, sample rate, free memory, connected browsers and similar.
- **Performance and heat:** chip temperature (shown as throttling when hot), analysis time per cycle, free memory, browsers connected, FIFO overflows, I2C errors and mesh nodes heard.



Licence and firmware

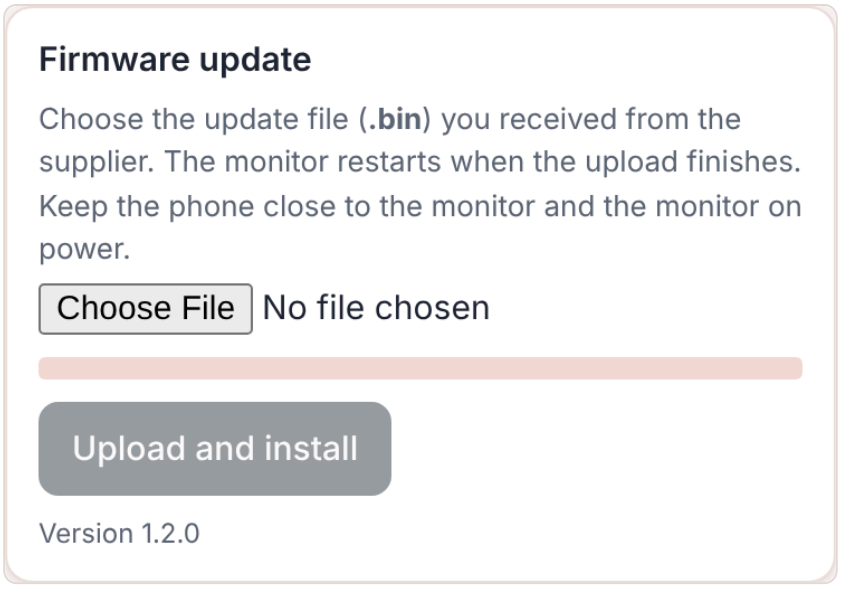
Licence



- **Status:** active, expired, not activated, wrong device or invalid.
- **Device ID (MAC):** give this to the designer or enter it in the key generator. A key only works on the monitor whose device ID it was made for.
- **Licensed to, Issued, Expires:** from the key.
- **Clock:** expiry is checked only once the clock is known (from your phone's time, or the internet when joined to a router).

To activate: paste the key into **Activation key** and press **Activate**. A message tells you whether it was accepted. **Copy device ID (MAC)** copies the MAC. **Remove key** deletes the key and returns to basic mode.

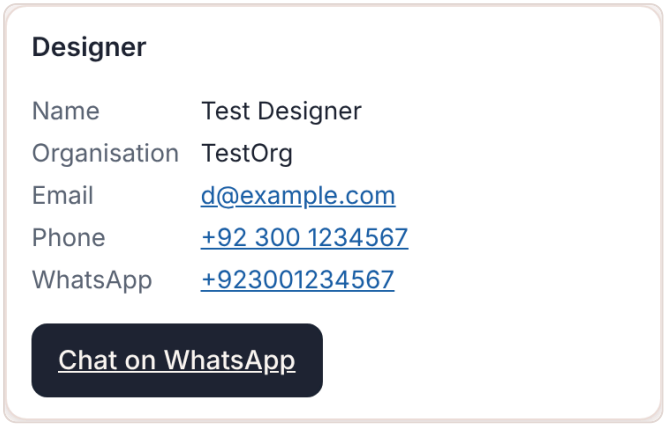
Firmware update

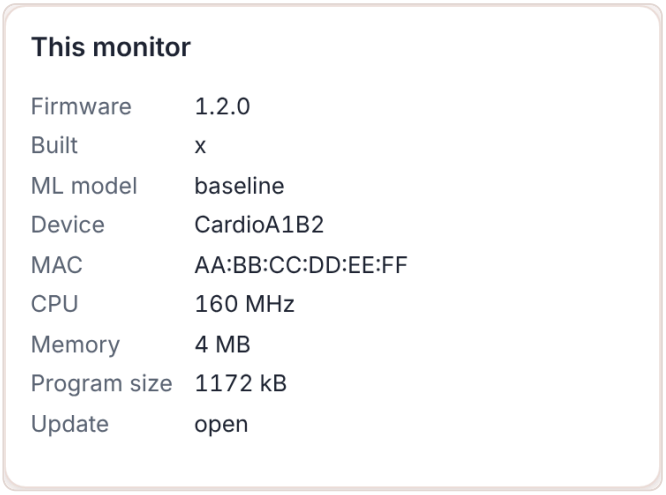


1. Press the file chooser and select the update file (`.bin`) you received from the supplier.
2. If the monitor uses an update password, type it in **Update password**.
3. Press **Upload and install**. The bar shows progress.
4. When it says "Installed" the monitor restarts; the page reconnects by itself after about 12 seconds.

> **Do not switch off** the monitor or leave the Wi-Fi range during the update. Keep the monitor on mains power or a charged battery.

About (bottom of the Settings page)

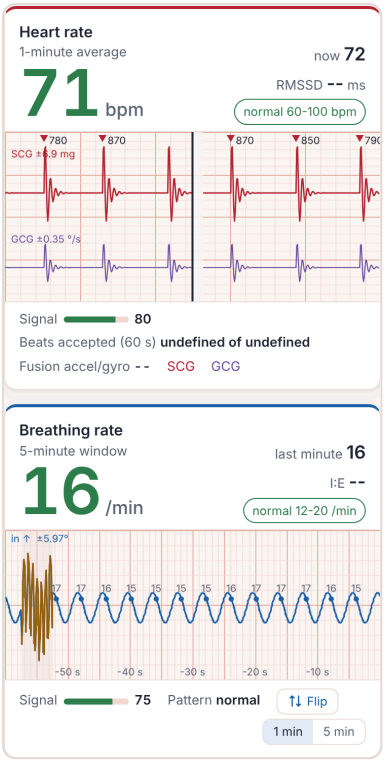




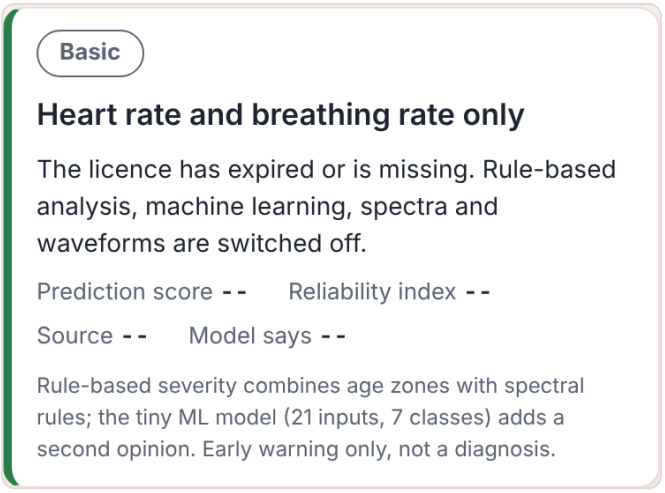
- **Designer:** name, organisation, email (tap to write), phone (tap to call) and WhatsApp.
- **Chat on WhatsApp** opens a chat with the designer, for example to ask for support or a new licence.
- **This monitor:** software version, build date, model version, device name, device ID and update mode.

15. Licence and the basic (expired) mode

State	What you get
Active licence	everything: waveforms, spectra, analysis, rule-based severity, machine learning, reports, mesh severity, full recordings
Expired or missing licence	**Basic mode:** only heart rate and breathing rate (with their range colours). Waveforms, spectra, rules, machine learning, the placement card and the interpretation are switched off. The breathing-pause, heart-rate, breathing-rate, temperature, sensor and connection alarms keep working. The assessment card says "Heart rate and breathing rate only".

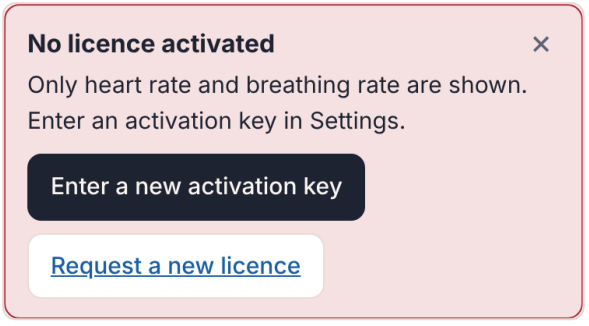


Basic mode: heart rate and breathing rate only.



The assessment card is replaced by a notice.

The licence banner



When the licence is missing or expired a banner appears on every page:

- **Enter a new activation key** jumps to Settings and puts the cursor in the key box.

- **Request a new licence** opens WhatsApp with a ready message that includes this monitor's MAC address.

Seven days before expiry an amber banner "Licence ends in N day(s)" appears, so you can renew in time.

How to get and enter a new licence

1. Send the monitor's **Device ID (MAC)** (Settings, Licence) to the designer using the banner button or the About card at the bottom of Settings.
2. The designer creates an activation key for that MAC and an expiry date.
3. Paste the key into **Activation key** and press **Activate**. The screen leaves basic mode within a second.

> **Note.** Changing the phone's date backwards does not extend a licence; the monitor remembers the latest time it has seen.

16. Heat, power modes and battery

The monitor's control unit is small and warms up mostly because of its Wi-Fi radio and processor. Version 1.2 (final) therefore starts in **Cool** mode (80 MHz processor, low radio power, slower waveform stream) and lets you change it in **Settings → Power and heat mode** (chapter 14).

What the firmware does	Effect
80 MHz clock in Cool and Eco	roughly a third less processor heat than 160 MHz
Lower Wi-Fi transmit power (5 dBm Cool, 2 dBm Eco)	less radio heat; keep the phone within a few metres
Slower waveform and spectrum streams	fewer Wi-Fi packets
Idle sleep when nobody is connected and nothing is recording	the processor rests most of the time
Thermal governor	above about 80 °C the clock stays at 80 MHz and the update rate is halved
Automatic boost	if the analysis takes too long or samples are lost, 160 MHz is used for up to 5 minutes, then it returns to the chosen mode

You can help further:

- Use **home Wi-Fi mode** instead of the hotspot when possible; it is clearly cooler.
- Keep the mesh **off** when not needed (Mesh page).
- Keep only one phone connected; each browser adds radio traffic. Close the dashboard when you are not looking at it.
- Do not cover the control unit or wrap it in foam or cloth; keep it out of the sun and away from the body. Only the sensor needs to touch the skin.
- Use **Eco** for long recordings. Check **Settings → Performance and heat** for the chip temperature.

> **Honest note.** The control unit will always feel warm, because the processor and the Wi-Fi radio are packed into a very small space. Feeling warm to the touch is normal. Cool and Eco reduce it but cannot make it cold. If it feels hot, choose Eco and contact your supplier if it stays hot.

Battery: monitors with a battery show its charge in the top bar; on mains power the bar shows "USB power".

17. Troubleshooting

Symptom	What to do
Page does not open	Join the monitor's network (`CardioResp-XXXXXX`) or your home Wi-Fi and open `http://192.168.4.1` or the saved address. On Android use the IP address instead of `.local`.
Phone keeps leaving the network	Stay connected despite "no internet", turn off mobile data, stay within a few metres.
Top bar shows "Reconnecting"	Out of range, or the monitor restarted. The page reconnects by itself.
Heart rate shows --	Press the sensor firmly, try the lower breastbone, keep still, wait 20 seconds.
Breathing weak when lying down	Move the sensor to the lower sternum or lower ribcage, then press **Recalibrate sensor** (Settings → Calibration).
A message keeps coming back after closing it	Its content changed (a new or different message). Read it; it closes again with ×.
Many false alarms	Check the **Reliability index** and signal quality; re-fix the sensor; confirm the **age** is correct; recalibrate.
Temperature seems wrong	Calibrate it against a thermometer (Settings).
"Basic" mode unexpectedly	The licence is missing, expired or for another monitor. See Settings → Licence.
Activation key rejected	Check that the key is complete (no missing characters), made for this monitor's MAC and not expired.
Cannot record	"measurement is not running": wait for calibration. "Not enough memory": delete old recordings.
Firmware update fails	Use the update file from your supplier, enter the update password if one is set, keep the phone close and try again. If the monitor does not start afterwards, contact your supplier.
Mesh shows nothing	Mesh on, the same group name and PIN on every monitor, all on the same Wi-Fi channel; press **Verify nearby monitors** (chapter 14).
A monitor is listed as "wrong PIN"	Enter the same PIN on both monitors and save.
Download does nothing	Open the address in the normal browser, not the Wi-Fi sign-in window.
Sensor not found / connection lost	Check that the sensor cable is plugged in firmly and not pulled or bent; switch the monitor off and on.
The traces look cut off or flat during movement	they are no longer blanked. During movement the scale grows and the movement is marked in amber; rates simply skip the moving seconds. If a trace is really flat, check the sensor contact and its cable.
Placement card: the dot is far from the centre while lying flat	the sensor is not mounted with Z out of the chest; re-mount it, then recalibrate (Settings → Calibration).
No alarm sound	Tap the page once; check **Alarms on** in the toolbar, the volume in Settings and the phone's media volume; use the Test buttons; keep the

	page open and the screen on.
Alarm keeps sounding	Go to the patient first. Then press **Silence 2 min**. Check the sensor contact and the age; a temperature alarm needs the temperature to be calibrated.
The monitor is hot	Settings → Power and heat mode → Eco; see chapter 16.

18. Quick reference and glossary

Quick start in six steps

1. Fix the sensor firmly on the lower breastbone.
2. Power the monitor; keep still for 8 seconds while it calibrates.
3. Join the monitor's network and open `http://192.168.4.1`.
4. People page: enter the **age** and press **Apply age**.
5. Live page: check the placement card, tap the page once so alarms can sound, then read the heart rate, breathing rate, assessment and colours.
6. Live page: **Capture 10 min** and **PDF report**; or Files page: record 1 minute and download its **PDF**. Settings: enter the licence key if needed.

Glossary

Term	Meaning
SCG / GCG	seismocardiogram / gyrocardiogram: chest vibration / rotation caused by the heartbeat
bpm	heart beats per minute
RMSSD, SDNN, pNN50	measures of beat-to-beat variability
I:E	inhale to exhale time ratio
Apnoea	a pause in breathing
Prediction score	model probability for its top class
Reliability index	how much to trust the current result (0-100)
mg	thousandths of the acceleration of gravity
Device ID (MAC)	the monitor's unique hardware address, used to bind the licence
Gravity reference	the up direction measured during the 8-second start-up calibration; tilt is measured against it
Mesh	several monitors sharing their patients by short-range radio, protected by a group name and PIN
.crr	raw recording file format of the monitor
Average heartbeat	the chest vibration of many beats laid on top of each other and averaged

Limits you should know

- Seismocardiography measures mechanical heart activity. It cannot diagnose arrhythmias, a heart attack or an infection; confirm with an ECG and a clinician.
- Oxygen saturation is not measured.
- Breathing depth is a relative tilt angle, not a volume.

- Accuracy depends on firm contact, placement and stillness. Compare against a reference device before relying on it.
- The shipped machine-learning model is a baseline; retrain it with real data for serious use.

Support: see the **About** card at the bottom of the Settings page for the designer's email, phone and WhatsApp.